Decomposing Illumination-Induced Variability in UAV Hyperspectral Canopy Reflectance Improves Leaf Chlorophyll Content Estimation

Yangming Feng, Puchen Yan, Jing M. Chen, Lei Gao, Yongyuan Gao, Rongpeng Zhang, Dalei Hao, and Shaozhong Kang

***Abstract*—Accurate estimation of leaf chlorophyll content (LCC) from centimeter-level uncrewed aerial vehicle (UAV) hyperspectral imagery is often challenged by illumination-induced variability within crop canopies, particularly in row crops where directional illumination interacts with complex canopy architecture. This variability reshapes the distribution of within-canopy reflectance and introduces spectral mixing that confounds biophysical retrieval. This study addresses this issue by decomposing the within-canopy reflectance distribution using a Gaussian mixture model (GMM), enabling the separation of illumination-consistent reflectance components under heterogeneous lighting conditions. The component with the highest mean reflectance was selected to represent canopy spectra less affected by shadow-induced mixing. LCC was estimated using vegetation indices (VIs), an inverted radiative transfer model (PROSAIL), and partial least squares regression (PLSR). The results show that the selected high-reflectance component consistently improves LCC estimation relative to canopy-averaged reflectance. In PROSAIL-based inversion, the coefficient of determination (R^2) increased from 0.67 to 0.84, while relative RMSE decreased from 16.91% to 10.17% (a 40% reduction). Across six VIs, the selected component improved VI-LCC relationships relative to binary sunlit-shaded thresholding ($\Delta R^2 \approx$ 0.051–0.118) and increased R^2 by 0.063–0.177 compared with using all vegetation pixels. In data-driven PLSR modeling, the component-based input raised the validation R^2 from 0.46 to 0.56 and reduced RMSE from 2.23 to 2.03 μg·cm^{-2} (double-split means), with consistent improvements confirmed under repeated double-split validation and predicted residual error sum of squares (PRESS)-based component selection. These results indicate that decomposing illumination-induced reflectance variability enhances the robustness of LCC estimation from UAV hyperspectral canopy reflectance under heterogeneous illumination conditions.**

***Index Terms*—Canopy structure, Gaussian mixture model (GMM), hyperspectral imagery, leaf chlorophyll content (LCC), solar angle, uncrewed aerial vehicle (UAV).**

I. Introduction

CHLOROPHYLL molecules convert light energy into chemical energy and are a key indicator of photosynthetic capacity [1]. In precision agriculture, timely monitoring of leaf chlorophyll content (LCC) can help farmers adjust irrigation and fertilization amounts to improve crop yield, as LCC is highly correlated with leaf nitrogen content and water conditions [2], [3]. Nondestructive and efficient monitoring of LCC based on optical signals opens a new avenue for assessing the physiological traits, growth status, and ecological functions of vegetation. However, traditional ground-based and satellite-based remote sensing methods are limited by insufficient temporal resolution, spatial resolution, and spatial coverage, and therefore cannot meet the high-precision requirements for canopy LCC estimation [4]. In recent years, the rapid development of uncrewed aerial vehicle (UAV) remote sensing technology has provided a new approach for high-precision estimation of LCC. UAV hyperspectral remote sensing offers high spatial resolution and operational flexibility, enabling detailed acquisition of canopy spectral information for LCC estimation. Notably, at centimeter to subcentimeter resolution, the pixel size approaches or even falls below the characteristic size of individual leaves, so that pixel reflectance no longer represents a spatially averaged signal but instead exhibits a continuous distribution rather than a simple binary (sunlit/shaded) classification [5], [6]. This continuity is driven by continuously varying leaf inclination angles and within-canopy leaf positioning, which create a continuum of local illumination states as solar geometry changes [7]. However, many existing analytical strategies still rely on binary or coarse categorical representations [8].

UAV-based LCC retrieval relies on canopy reflectance factors derived from radiometrically calibrated imagery. In practice, UAV data acquisition is flexible, with high freedom in flight time and location, so the data are often not under the

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Yangming Feng, Rongpeng Zhang, and Shaozhong Kang are with the State Key Laboratory of Efficient Utilization of Agricultural Water Resources, China Agricultural University, Beijing 100083, China (e-mail: kangsz@cau.edu.cn).

Puchen Yan is with the National Field Scientific Observation and Research Station on Efficient Water Use of Oasis Agriculture, China Agricultural University, Beijing 100083, China.

Jing M. Chen is with the Department of Geography and Planning, University of Toronto, Toronto, ON M5S 3G3, Canada.

Lei Gao is with the Center for Agricultural Water Research in China, China Agricultural University, Beijing 100083, China.

Yongyuan Gao is with the College of Land Science and Technology, China Agricultural University, Beijing 100083, China.

Dalei Hao is with the Atmospheric, Climate, and Earth Sciences Division, Pacific Northwest National Laboratory, Richland, WA 99352 USA.

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same illumination conditions. This affects the consistency of the canopy reflectance data, also limiting the lateral comparison of canopy reflectance spectral images acquired by UAVs [6]. Even if data are acquired at a fixed location and a fixed time window, the solar angle still changes daily due to the Earth's rotation, which is often overlooked. Since reflectance is the foundation of optical remote sensing, uncertainties can propagate through canopy reflectance to the calculation of vegetation indices (VIs) and the estimation of vegetation traits [5], [9], [10]. Furthermore, the recent surge in data-driven methods has placed high demands on data quality, but variability caused by different illumination conditions [e.g., solar zenith angle (SZA) and aerosol optical depth] can lead to problematic results in data-driven approaches.

Two main factors influence the canopy reflectance data acquired by UAVs: the first is the influence of sun-target-observation geometry [11], [12]. The anisotropy of solar irradiance is the fundamental cause of directional illumination issues. Typically, under clear sky conditions dominated by direct sunlight, occlusion by branches and leaves within the canopy creates sunlit and shaded leaf portions. Due to the nonuniform distribution of foliage, the proportion and reflected radiance intensity of these sunlit and shaded components vary with differing solar illumination angles and sensor viewing angles. For instance, the "hotspot effect"—a surge in canopy reflectance—occurs when the sensor's viewing direction nearly aligns with the solar incidence direction [13]. The bidirectional reflectance distribution function (BRDF) studies clearly demonstrate that canopy reflectance varies with illumination and observation angles, generally showing minimum reflectance when viewed opposite to the illumination direction and maximum reflectance when viewed from the same direction as the illumination, primarily due to changes in the proportion of mutually shaded leaves [14]. However, the application of comprehensive BRDF correction typically requires multiangular observations, which are not always feasible or obtainable in standard single-view UAV data acquisition, thereby limiting their direct utility in mitigating angular-induced reflectance variability in such scenarios. Similar angular patterns are observed in vegetation parameters derived from reflectance [15], although the specific reflectance distribution can vary significantly across different canopy types [16]. This is due to the second factor: the complexity of canopy structure [7]. Because the distribution of leaves between canopies and within canopies is not ideal, the position, proportion, and reflected radiation of sunlit and shaded leaves change when observed from different angles. For example, the SAIL model quantifies the nonlinear increase in canopy reflectance with increasing leaf area index (LAI) and the phenomenon of near-infrared (NIR) saturation [17]. Although models like SAIL provide valuable insights, they typically assume a homogeneous "turbid medium" canopy, which may not adequately represent the complex structural heterogeneity captured by centimeter-level UAV imagery, and thus may oversimplify the actual light interactions within complex canopies. The clumping effect can create more shadows, thereby affecting canopy reflectance [18], and may cause a decrease in red light reflectance and an increase in NIR reflectance [19]. Studies on crop leaf angle distribution (LAD) have also found that in the red band, the canopy reflectance of horizontally distributed LAD is lower than that of erectophile canopies, but higher in the NIR band than that of erectophile LAD canopies [20]. For crops like cotton and corn, the influence of row structure cannot be ignored [10], [11].

To address these reflectance variations, previous research has explored several strategies to suppress or reduce their impact, many of which are represented in the Supplementary Material. These strategies can be broadly categorized. First, a substantial body of work has focused on the direct characterization and modeling of the BRDF. This involves understanding how reflectance changes with viewing and illumination geometry. Early geometric-optical models (e.g., [21]) treated canopies as collections of discrete objects (e.g., crowns), explicitly modeling how crown shape, density, and mutual shadowing control directional reflectance, including the hotspot effect. Later, semi-empirical approaches, notably linear kernel-driven models (e.g., [22], [23]), became widely used. These models deconstruct the BRDF into a weighted sum of mathematical functions, or "kernels," each representing a scattering mechanism (e.g., volumetric scattering from leaf layers and geometric-optical scattering from surface structural elements). By fitting these models to multiangular observations, canopy reflectance can be normalized to a standard sun-target-sensor geometry, thereby mitigating variations. Such BRDF modeling has also been applied to normalize angular effects in canopy reflectance and solar-induced fluorescence (SIF) [24]. Qin et al. [25] proposed hemisphere harmonics (HSHs) basis functions as a universal BRDF representation that avoids landcover-specific structural assumptions, validated on UAV hyperspectral data across six landcover types.

Second, another common strategy, enabled by high-resolution imagery, is to separate pixels within the canopy into sunlit and shaded components. The core idea is to isolate pixel populations that better represent vegetation optical properties by reducing confounding effects from shadows, varying illumination, or soil background. This includes methods that explicitly model and separate sunlit and shaded leaf fractions based on canopy structure (e.g., LAI and clumping index) and view/illumination geometry, often aiming to normalize the signal to the sunlit component or derive a weighted total canopy response [26]. Other techniques use the statistical distribution of pixel apparent reflectance in high-resolution imagery (e.g., in the NIR band) to partition pixels into sunlit, shaded, and problematic (e.g., specular, deep shadow, and background) categories, thereby focusing subsequent analysis on the most informative pixels [27]. In most of these approaches, Gaussian fitting is primarily used to derive global apparent reflectance thresholds for binary or quasi-binary pixel partitioning, rather than to assign posterior illumination labels at the pixel level. Furthermore, some approaches classify vegetation pixels and then select specific apparent reflectance fractions—such as the highest-reflectance green pixels to enhance signal from well-illuminated leaves [28], or even the lowest-reflectance green pixels to mitigate high-reflectance artifacts associated with leaf orientation and pixel-scale misregistration when

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estimating leaf biochemistry [4]. Ihalainen et al. [29] took a physically based route, recovering true leaf reflectance from apparent top-of-canopy spectra by estimating five spectral-invariant parameters with a random-forest regressor trained on Monte Carlo ray-tracing simulations. At the mosaic level, Xie et al. [30] decomposed UAV hyperspectral strips into reflectance and shading components to homogenize illumination across adjacent strips during stitching.

Third, considerable effort has been directed toward constructing VIs that are less sensitive to confounding factors, such as LAI, solar angle, or canopy architectural effects like row orientation, while maximizing sensitivity to the target biophysical parameter (e.g., LCC). This is often achieved by combining reflectance from multiple spectral bands. Some VIs, such as the Angular Insensitivity Vegetation Index (AIVI) [31] or the LAI-Insensitive Chlorophyll Index (LICI) [32], are formulated by mathematically combining simpler band ratios or indices that exhibit differential sensitivities to the target and confounding variables. Other VIs, such as the Urban Tree Chlorophyll Index (UTCI) [27] or the mND_{blue} [28], are designed by selecting narrow spectral bands (often in the red-edge or using blue reference bands) that are more robust to structural or illumination effects due to their position relative to pigment absorption features or atmospheric windows. More recently, spectral transforms (e.g., fractional-order derivatives and wavelets) and multisource image fusion have been combined with multiband index construction to improve leaf-trait retrieval [33], [34], [35]. The development of such indices often involves systematic testing across diverse conditions to identify band combinations and formulations with improved stability and accuracy. Complementing VI design, optimizing observation timing (e.g., off-noon for row crops) has also been explored to minimize influences such as sunlit soil visibility [32]. Although these methods have advanced the field, many approaches face challenges when applied to high-resolution, single-view UAV imagery, often requiring substantial ancillary data or simplified assumptions that cannot fully capture the observed reflectance variability. See Table S1 in the Supplementary Material for a representative summary.

Taken together, the above strategies either normalize angular effects through explicit BRDF characterization, or reduce their influence by selecting representative pixel subsets or designing indices that are relatively robust to confounding factors. More fundamentally, an unresolved question is whether illumination-aware reflectance representations can yield consistent improvements in LCC retrieval across fundamentally different modeling paradigms, rather than being effective only within a specific inversion or regression framework. However, these approaches often implicitly assume that within-canopy illumination variability can be adequately summarized by a small number of discrete classes, or by globally defined thresholds derived from the reflectance distribution. This assumption becomes increasingly fragile for centimeter-level UAV hyperspectral imagery acquired under a single-view geometry, where the apparent reflectance of vegetation pixels commonly exhibits a continuous distribution, whose shape can vary with solar geometry. In particular, as the SZA increases, the reflectance distribution may depart from a simple bimodal form and develop more complex features such as skewness and heavy tails, making dichotomous or threshold-based partitioning less reliable for consistently isolating leaf-optical signals. In this study, both field observations and large-scale emulation system (LESS) simulations provide quantitative evidence for this distributional reshaping, motivating the adoption of a pixelwise, distribution-aware modeling framework.

Building on this observation, this article proposes a Gaussian mixture model (GMM)-based framework for estimating LCC from pixel-level apparent reflectance. The main contributions are as follows.

1) We introduce a pixelwise, posterior-based GMM framework to represent illumination-driven components within continuous canopy reflectance distributions, avoiding predefined sunlit–shaded thresholds.
2) We adapt this framework to single-view nadir UAV hyperspectral imagery, requiring only standard acquisition geometry and remaining effective under varying solar angles.
3) We demonstrate that selecting the highest-mean reflectance component consistently improves LCC retrieval across VI-based regression, physics-based PROSAIL inversion, and partial least squares regression (PLSR).

In this study, cotton canopies exhibited heterogeneous structures due to water and salinity stress, providing a representative test case for evaluating illumination effects. Concurrently, canopy reflectance spectra under various conditions were simulated using the LESS and PROSPECT. We investigated the influence of different SZAs and azimuth angles on canopy reflectance spectra and proposed an analysis framework based on the continuous probability distribution features of image reflectance. Based on the above motivation, the specific aims of this study are: 1) to examine canopy-reflectance variation under different solar zenith and azimuth angles and canopy structures and 2) to identify reflectance components within canopy images that more consistently represent canopy status under varying solar geometry, and to use these components to improve the accuracy and stability of LCC estimation. This work provides a distribution-aware approach for using UAV-based vegetation remote sensing spectra.

II. Materials and Methods

A. Field Experiments Data

1) UAV Data Acquisition: Field measurements were conducted in 2022 and 2023 at two experimental sites located in Aral, Xinjiang, China (81; 12°E, 40; 24°N, altitude 957.4 m). Both sites have similar soil types (sandy loam soil) and were planted with cotton (Tahe 2). The 2022 site covered an area of approximately 1500 m^2, and the 2023 site approximately 3000 m^2; both were elongated in shape, with sample plots concentrated within a central strip of 20–25 m width. Aral is situated in the Tarim Basin of southern Xinjiang—one of China's primary cotton-producing regions—where soil salinization and water stress are widespread, providing a representative setting for studying illumination effects on LCC retrieval under realistic field conditions [36]. The geographic

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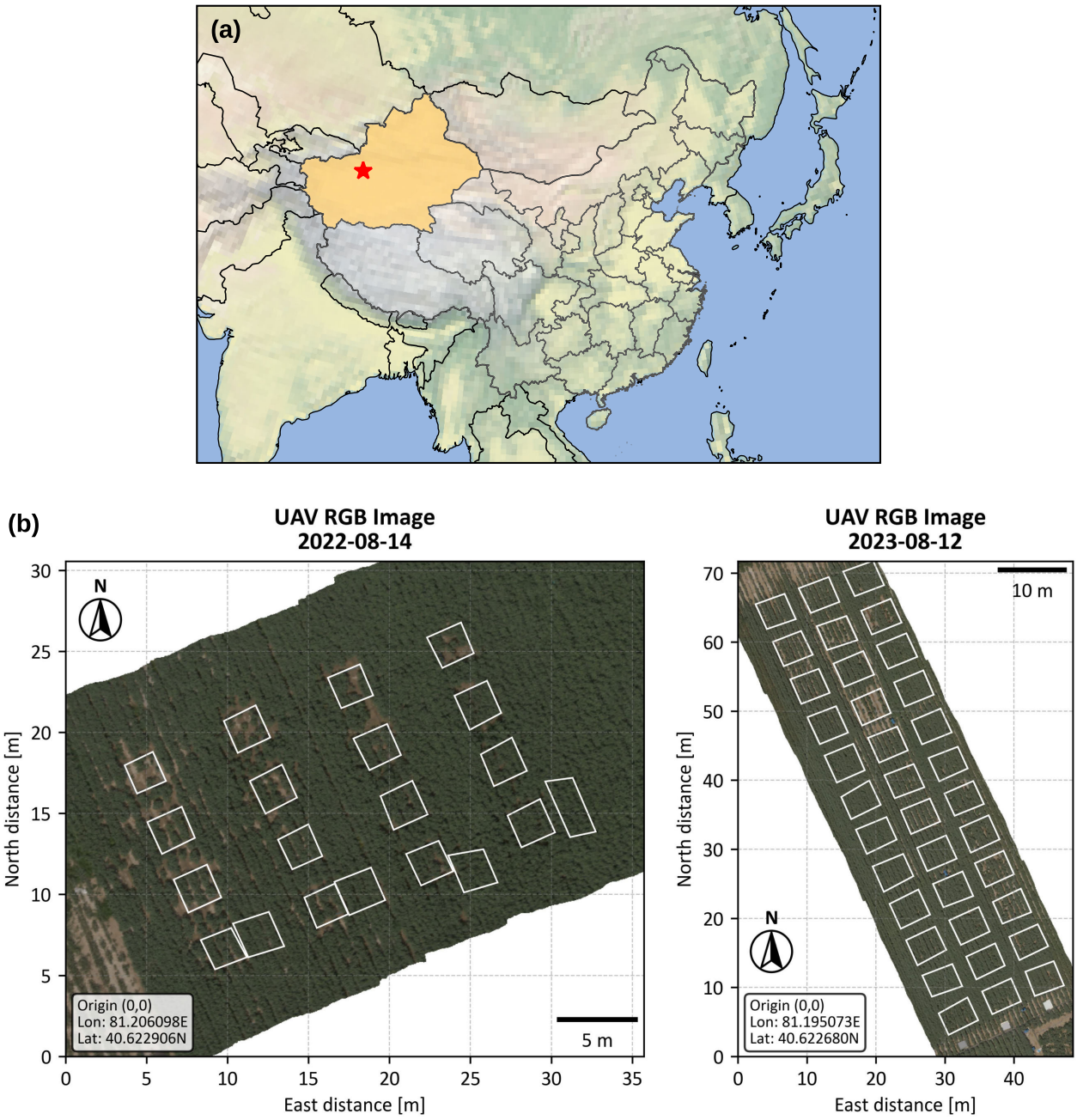


Fig. 1. Geographic location and field layout of the experimental sites. (a) Location of the study area (red star) within Xinjiang Autonomous Region (highlighted in orange), China. (b) Spatial distribution of experimental plots overlaid on UAV RGB imagery: left panel shows the 2022 campaign (August 14, 2022) with 2×2 m ROIs; right panel shows the 2023 campaign (August 12, 2023) with 4×4 m ROIs. White rectangles indicate the pixel-extraction areas used for spectral analysis. North arrows and scale bars are provided for reference.

location of the study area and the spatial distribution of sample plots are shown in Fig. 1. Except for irrigation, all other agronomic practices followed local multiyear planting conventions to ensure crop growth. We prioritized flying along the longest dimension of the field to minimize the number of turns during operation, thereby enhancing data acquisition efficiency and ensuring complete coverage by the flight path. Concurrently, the flight line design ensured that each experimental plot was fully captured within one or two adjacent flight swaths, with flight lines oriented as parallel or perpendicular as possible to plot boundaries to facilitate subsequent data processing and precise extraction of regions of interest. Based on these principles, in 2022, the experimental field was longer from north to south and

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TABLE I
SUMMARY OF SOLAR AND WEATHER INFORMATION DURING HSI DATA ACQUISITION. THE TIME ZONE USED IN ARAL IS BEIJING TIME (UTC+ 8)

Date	Zenith [°]	Azimuth [°]	Sky condition
2022			
5 July	18.17	192.13	Sunny
9 July	24.61	227.96	Cloudy
12 July	19.76	158.42	Sunny
13 July	22.23	216.44	Cloudy
16 July	19.30	176.29	Sunny
20 July	20.21	170.10	Sunny
23 July	20.56	178.87	Sunny
27 July	20.56	178.87	Sunny
2023			
15 July	19.08	182.74	Sunny
18 July	20.49	199.42	Sunny
9 Aug.	28.81	215.83	Sunny
9 Aug.	36.68	236.78	Sunny
9 Aug.	46.60	251.64	Sunny
9 Aug.	58.08	263.69	Sunny
9 Aug.	67.74	272.19	Sunny
9 Aug.	80.28	282.66	Sunny

narrower from east to west, with experimental plots arranged in north–south strips. Consequently, the flight lines were set to an east–west direction, resulting in the flight direction being perpendicular to the crop row direction (the measured row orientation was approximately 158° that year). In 2023, the field was longer from east to west and narrower from north to south. Therefore, the flight lines were adjusted to a north–south direction, making the flight direction parallel to the crop row direction (the measured row orientation was approximately 156° that year). The actual crop row orientation remained largely consistent between the two years. The potential impact of crop row structure on canopy reflectance is discussed in Section IV-D. Table I shows the date, solar angle, and weather information for the measurements.

The hyperspectral system used in this study is a line-scanning hyperspectral camera, Pika L (Resonon Inc., Bozeman, MT, USA), which acquires images using a line-scanning imaging method. This imaging method mainly employs nadir viewing geometry. The hyperspectral camera is mounted on a gimbal equipped with a Matrice 600 Pro (DJI Technology Company Ltd., Shenzhen, China) to reduce the vibrations of the hyperspectral sensor caused by aircraft jitter and braking inertia during flight. The Pika L has 281 spectral channels covering the spectral range of 400–1000 nm with an average FWHM of 2.7 nm, 900 spatial pixels in the vertical flight direction, and is equipped with objective lenses with a focal length of 17 mm, an FOV of 17.6°, and an IFOV of 0.71 mrad. In each flight, the lens of the Pika L was set to vertically downward observation, and the line-scanning direction was consistent with the UAV flight direction. The UAV flight altitude was set to 80 m above ground level, and the flight speed was set to 3 m/s. Therefore, the nominal spatial resolution of the acquired hyperspectral images was 2.75 cm/pixel; the images were resampled to a uniform 4.00 cm/pixel, a value chosen to conservatively exceed the maximum instantaneous ground sampling distance arising from real-time variations in flight altitude and airspeed during push-broom acquisition, thereby ensuring spatial homogeneity of the mosaicked dataset.

Before the flight, three calibrated reference panels (each measuring 1.5 × 1.5 m) with reflectance of 30%, 50%, and 70% were placed on the ground within the flight path for radiometric correction. The reference panels were placed as far away from surrounding objects as possible, or elevated above them. A single flight was controlled to be completed within 10 min to ensure that the illumination conditions did not change dramatically during the flight, even in cloudy weather. During the flight, a USB 2000 + (Ocean Insight, USA) spectroradiometer and an integrating sphere were used to continuously monitor changes in the total incoming, direct incoming, and diffuse incoming irradiance spectra in the vertical direction. This ground-based instrument was used to characterize ambient illumination conditions, specifically for deriving the skylight proportion (*skyl*) as an input for the LESS model and to aid in interpreting canopy reflectance variations. This was distinct from the radiometric calibration of the UAV hyperspectral data, which was performed using ground-deployed reference panels and an empirical line method.

2) Leaf and Canopy Data Collection: Ground data were collected on the same day as, or the day after, each flight. In 2022, 20 plots of 2 × 2 m were surveyed, and in 2023, 36 plots of 4 × 4 m were surveyed. Plot size differed between years due to site changes. The larger 4 × 4 m plots were adopted in 2023 at the new site to account for potential unknown micro-environmental variability and to further ensure the representativeness of the core sampling area by minimizing edge effects. These plots were kept away from the field edge to reduce plant-community edge effects. The plots were distributed evenly across the entire planting area: in 2022, the 2 × 2 m plots were spaced 2 m apart in the north–south direction and 4 m apart in the east–west direction; in 2023, the 4 × 4 m plots were spaced 2 m apart in both directions [see Fig. 1(b)], thus ensuring the representativeness of the ground data and the reference for comparison with the UAV data. The geographic coordinates of each plot center were recorded in the field, and each plot was further marked with yellow labeled stakes bearing the plot identifier code together with blue ground markers, positioned to be identifiable in the UAV imagery while minimizing interference with plant growth. These markers are clearly discernible in the centimeter-resolution UAV imagery, and the regions of interest were delineated by identifying them in the georeferenced images, in agreement with the recorded plot-center coordinates, ensuring that the pixels extracted for each region of interest correspond to the plants sampled in the field. The collection of ground data included soil plant analysis development (SPAD) and leaf reflectance.

SPAD, LCC, and leaf reflectance were measured from fresh leaves collected in the field and transported to the laboratory. To ensure representative and consistent sampling for SPAD measurements, three typical cotton plants exhibiting uniform growth and free from disease or pests were selected from each plot to represent the within-plot spatial variability. From each selected plant, after discarding newly unfurled apical leaves, the third fully expanded leaf from the apex of the main stem

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was chosen. For each such leaf, SPAD readings were taken using a SPAD-502Plus (Konica Minolta, Inc., Tokyo, Japan) at three distinct locations (tip, middle, and base), avoiding the main veins. These three intraleaf readings were averaged to obtain a mean SPAD value for that leaf (representing the plant). Finally, the mean SPAD values from the three selected plants within that sampling interval were averaged to constitute one composite sample reading for that location. Leaf reflectance was measured using a USB 2000 + (Ocean Insight, USA) spectroradiometer, integrating sphere, halogen lamp, and a 98.49% diffuse reflectance reference panel (400–1000 nm) (Changhui Electronic Technology Company Ltd., Guangzhou, China), assuming a Lambertian surface. Before each measurement, the spectroradiometer was calibrated using a white reference plate, with a measurement range of 400–1000 nm. Photographs of the field sampling and laboratory measurement procedures are provided in Supplementary Fig. S1.

Studies have shown excellent correlations between SPAD and LCC [37], [38], [39], [40], so we used SPAD values as a proxy for LCC. To verify this assumption, we conducted a correlation analysis between SPAD and LCC. We randomly selected 40 cotton leaves, covering a range from light to dark green. In the laboratory, the SPAD value of a leaf was measured at three different locations, and the average value was taken as the SPAD value of the leaf. The LCC of the leaf was then immediately measured using the standard wet chemical method. The correlation analysis results between SPAD value and LCC are shown in Fig. 2. The results show that there is a significant linear correlation between SPAD value and LCC, with a coefficient of determination (R^2) of 0.81, a relative root mean square error (rRMSE) of 8.27%, and a p-value less than 0.001. Therefore, we used SPAD values as a proxy for LCC.

In addition, the acquisition of the cotton 3-D plant models for LESS was done by placing potted cotton in a $3 \times 3 \times 2.5$ m darkroom, which was equipped with a central turntable, a calibration plate of known size, a digital camera Canon EOS1500D (Canon Inc., Tokyo, Japan), and top supplemental LED lights. During image acquisition, the digital camera was positioned at two heights: 1) the top of the plant and 2) half of the plant's height (from soil to canopy top). From both positions, the camera lens was aimed at the plant's half-height point. 2-D photographs were taken every 5° of rotation, achieving full 360° coverage [41], [42]. 144 photographs were taken for each cotton plant. Subsequently, 3-D reconstruction of the cotton plant and generation of OBJ format files were completed in 3DF Zephyr software (3Dflow Srl, Verona, Italy).

B. Simulation Data

To simulate canopy reflectance images approximating those obtained in field experiments, LESS [43] was used to simulate various canopy scenarios. LESS is a 3-D radiative transfer model based on ray tracing, capable of simulating remote sensing signals such as multispectral/hyperspectral images. It utilizes a weighted forward photon tracing method to simulate multispectral reflectance or radiation-related data, and a backward path tracing method to generate sensor images or large-scale spectral images. This study employed 3-D cotton

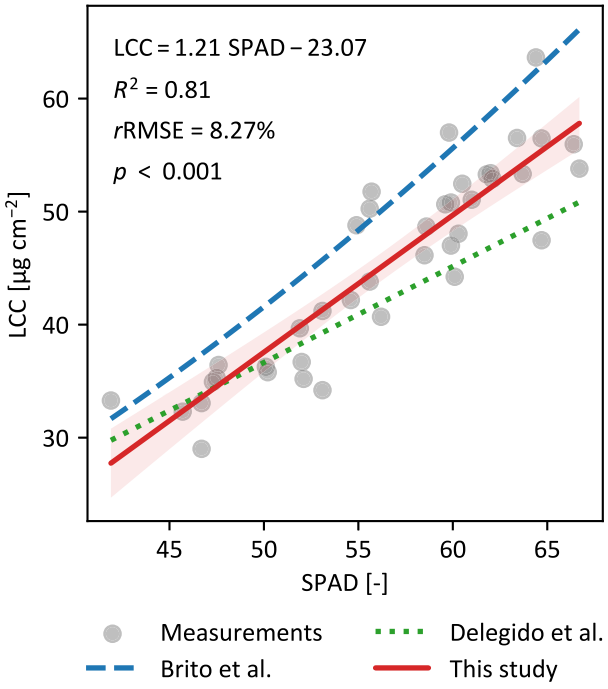


Fig. 2. Correlation between LCC proxy SPAD values and laboratory measured LCC [μg/cm^2] for cotton leaves. The red solid line represents the linear regression model of this study. The blue dashed line represents the regression model of Brito et al. [38]. The green dotted line represents the linear regression model of Delegido et al. [39]. The equations R^2, rRMSE, and p value of the linear regression model of this study are shown in the upper left corner.

plant models at four different growth stages. The distribution of the four 3-D plant models in the simulated scenarios strictly referenced the planting pattern in the field experiment, adopting a wide-narrow row planting format. Wide rows were spaced 0.66 m apart, narrow rows 0.10 m apart, and plants were spaced 0.10 m apart within the rows. The planting direction was 24° west of due north. The z-axis parameter of the plant models was set with a 5-cm random jitter to simulate the uneven height of crops in reality, and the rotation angle was randomly distributed between 0° and 360°. As cotton components other than leaves account for a small proportion, the default soil reflectance in LESS was used. The soil background in LESS was parameterized using locally measured soil reflectance spectra.

The optical properties of the four 3-D plant models were simulated using the PROSPECT-D model [44]. We used PROSPECT-D because the study targets LCC retrieval within 400–1000 nm, where PROSPECT-D is well validated. PROSPECT-PRO mainly benefits protein-related traits (via PROT and CBC separation) that are not targeted here and are better supported by SWIR information. Therefore, PROSPECT-D provides sufficient realism without adding unnecessary parameter complexity. The PROSPECT model [45] was initially designed to simulate leaf reflectance and transmittance, mainly by considering the effects of leaf structural parameters and biochemical components (such as chlorophyll and water content). With the iteration of versions, the PROSPECT model has continuously added new functions

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TABLE II
SUMMARY OF INPUT PARAMETERS FOR PROSPECT-D AND LESS MODELS

Model parameter	Variable [unit]	Range
PROSPECT-D		
Chlorophyll a+b concentration	C_{ab} [$\mu g/cm^2$]	10–80
Leaf structure parameter	N [−]	1–3
Carotenoid concentration	C_{car} [$\mu g/cm^2$]	Fixed at $C_{ab}/4$
Anthocyanin concentration	C_{anth} [$\mu g/cm^2$]	Fixed at 0
Brown pigment	C_{brown} [−]	Fixed at 0
Equivalent water thickness	C_w [cm]	Fixed at 0.03
Dry matter content	C_m [g/cm^2]	Fixed at 0.01
LESS		
Image dimension	*shape* [pixel × pixel]	Fixed at 250 × 250
Image resolution	*resolution* [cm/pixel]	Fixed at 4
LAI of scene	LAI [m^2 m^{-2}]	2.28, 4.60, 5.91, 8.34
VAA	VAA [°]	Fixed at 156
VZA	VZA [°]	Fixed at 0
Sky light percentage	*skyl* [%]	25%, 50%, 95%
SAA	SAA [°]	156, 186, 216, 246, 276, 306
SZA	SZA [°]	0, 20, 40, 60, 80

and improvements to enhance its applicability and accuracy in different vegetation types and environmental conditions.

Table II lists the parameter settings used to drive LESS and PROSPECT-D. LCC was represented in PROSPECT-D by the chlorophyll a + b parameter C_{ab}; hereafter, LCC refers to C_{ab} for consistency with the field measurements. Among the PROSPECT-D inputs, LCC was sampled from a truncated normal distribution (μ = 54, σ = 10), truncated to 10–80 $\mu g/cm^2$. The leaf structure parameter N was sampled from a truncated normal distribution (μ = 1.6, σ = 0.1), truncated to 1–3. Because carotenoid concentration is highly correlated with LCC [46], [47], C_{car} was set to LCC/4 [48]. Anthocyanin concentration and brown pigment were set to zero. The studied wavelength range is 400–1000 nm as the influence of LCC on canopy reflectance is mainly around 400–800 nm. Consequently, the influence of equivalent water thickness and dry matter content is very weak in this wavelength range [49]; thus, these two parameters were set to fixed values based on the laboratory measurements of cotton leaves. Among the parameters of the LESS model, the image dimension was fixed, i.e., 250 × 250 pixel. Considering that the image resolution is 4 cm/pixel, the scene size is 10×10 m. LAI is determined by the 3-D plant models of four different growth stages and their distribution, and the LAI within the 10×10 m scene is calculated. In order to be as close as possible to the real field experiment situation under the condition of ensuring the universality of the results, the observation azimuth angle was fixed at 156°, and the observation zenith angle was fixed at 0°. The solar azimuth angle was varied from 156° to 306° in 30° increments, and the SZA was varied from 0° to 80° in 20° increments. The diffuse fraction (*skyl*) was prescribed using representative values informed by ground-based measurements. To span practical UAV illumination conditions, three skyl levels (25%, 50%, and 95%) were simulated, representing predominantly sunny, cloudy, and highly diffuse illumination, respectively.

In total, we generated 5400 scenarios. Specifically, four growth stages, five SZAs, six SAAs, and three skyl levels resulted in 360 unique geometry–illumination combinations, and each combination was repeated 15 times with random draws of C_{ab} and N, together with random plant rotation and height jitter.

C. HSI Data Preprocessing

The measured data underwent geometric correction and radiometric calibration to generate reflectance images. Dark current is recorded by the onboard microcomputer, acquired before and after the flight, and interpolated for each frame (line) calculation during processing. Geometric correction of the HSI data is performed using MegaCube (LICA Company, Beijing, China) software, and the images are geographically registered in ArcGIS 10.8 (Esri, Inc., Redlands, California, USA) using ground control points (GCPs) to obtain georeferenced image cubes. Because the UAV operates at low altitude (80-m AGL), atmospheric effects are considered negligible in this study. We use the empirical line method to convert digital number (DN) value data into reflectance factor images [50]. The empirical line method is calculated as follows:

$$\rho(\lambda) = \mathrm{DN}_\lambda \cdot A_\lambda + B_\lambda \tag{1}$$

where $\rho(\lambda)$ is the reflectance at wavelength λ, DN_λ is the DN value at wavelength λ, and A_λ and B_λ are the slope and intercept, respectively. In this step, we used the reflectance measured in the laboratory for three reference panels deployed before the flight as the reference, and calculated A_λ and B_λ using a linear regression model. To keep the spectral resolution of the final experimental data consistent with that of the simulated data, the spectral resolution of the reflectance image was resampled to 1 nm.

Given the centimeter-level resolution of UAV imagery, soil background pixels can significantly bias the GMM-based separation of vegetation pixels with different apparent reflectance levels when the vegetation cover fraction (f_{Cover}) is low. Therefore, creating a mask using a normalized difference vegetation index (NDVI) threshold can effectively separate vegetation pixels, thereby eliminating the interference of soil pixels on the GMM fitting of the canopy probability density function (PDF). In this study, soil pixels were removed using a method based on calculating the NDVI threshold from the NDVI probability density function. Specific details of this method are provided in Supplementary Material. Although cotton leaves within a small area (1 × 1 m) are similar, the vegetation-pixel image still exhibits a range of apparent reflectance values, as shown in Fig. 3.

D. Pixel Clustering Based on GMM

To separate illumination-driven apparent reflectance subgroups within vegetation pixels, we fit the pixelwise apparent reflectance distribution using a GMM. Here, apparent

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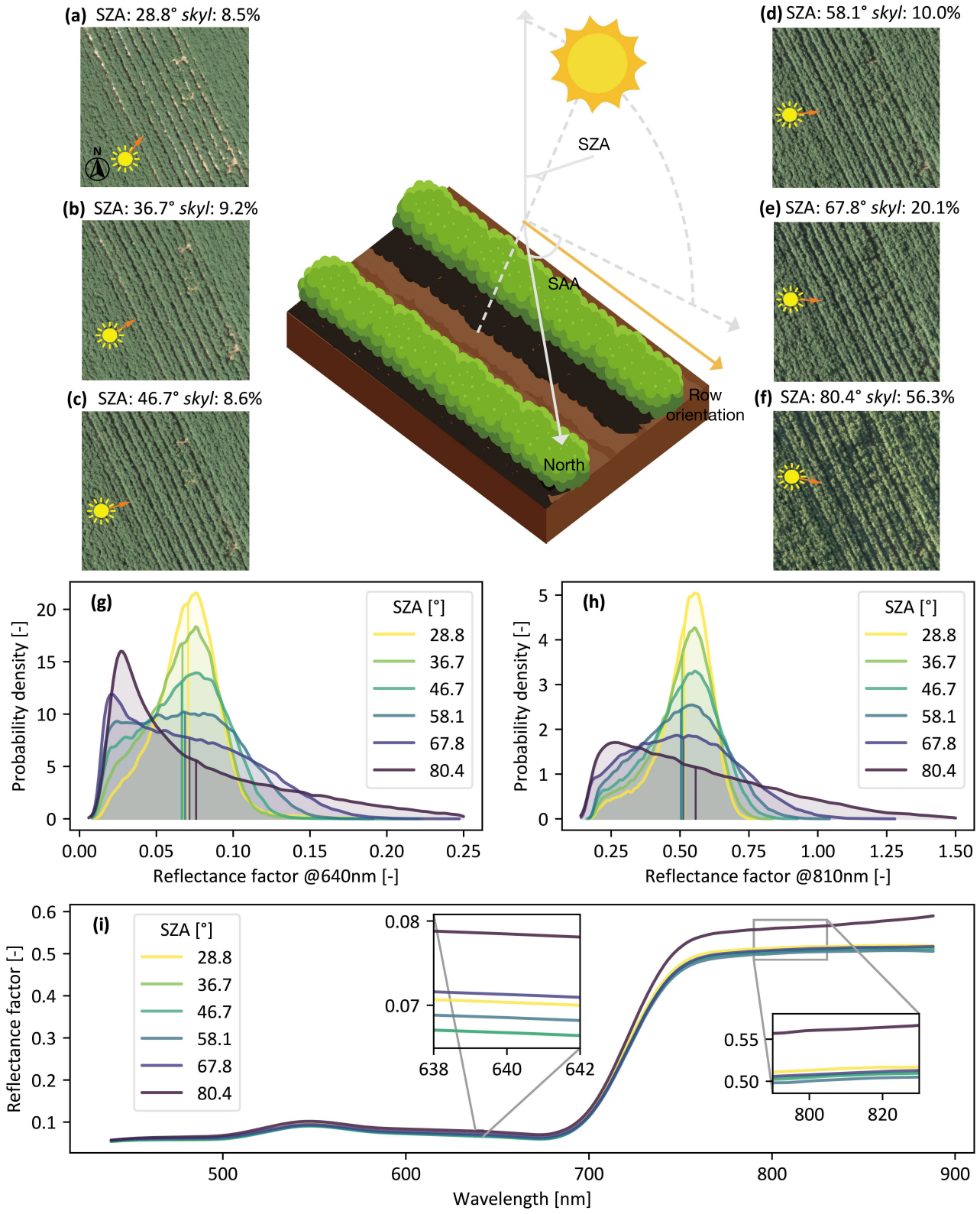


Fig. 3. Diurnal variation in reflectance factor and corresponding probability density functions (PDFs) for NIR and red wavelengths as a function of SZA. (a)–(f) Time series of UAV hyperspectral images acquired over the same cotton field [13 × 13 m], and solar azimuth angles are indicated by the sun locations and arrow icons in the images. PDFs of (g) NIR and (h) red reflectance factors at varying SZAs. (i) Hyperspectral reflectance factor from 400–900 nm at varying SZAs.

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reflectance refers to the observed pixel-level reflectance magnitude under a given sun-sensor geometry, reflecting both illumination and canopy-structure effects. The GMM is defined as

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k) \tag{2}$$

where x denotes the apparent reflectance metric of a vegetation pixel, K is the number of components, π_k is the mixture weight ($\sum_{k=1}^{K} \pi_k = 1$), and $\mathcal{N}(\cdot \mid \mu_k, \Sigma_k)$ is a Gaussian distribution with mean μ_k and covariance Σ_k.

Model parameters were estimated using the standard expectation–maximization (EM) algorithm, which yields posterior membership probabilities (responsibilities) for each pixel. The responsibility, i.e., the posterior probability that a pixel with apparent reflectance x belongs to component k, follows from Bayes' rule:

$$\gamma_k(x) = \frac{\pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(x \mid \mu_j, \Sigma_j)} \tag{3}$$

where π_k acts as the prior weight of component k and $\mathcal{N}(x \mid \mu_k, \Sigma_k)$ as its likelihood. We treat each component as an illumination-related reflectance group and perform pixelwise posterior labeling by assigning each vegetation pixel to the component with the maximum responsibility (i.e., a maximum a posteriori assignment), followed by componentwise aggregation for LCC retrieval. We evaluated K = 2–5 and report results across this range. The number of components is chosen per analysis rather than fixed, reflecting a tradeoff between resolving the within-canopy illumination gradient (a two-component split gives only a coarse sunlit–shaded contrast) and retaining enough pixels per component for stable statistics, which becomes more limiting as the pixel pool decreases. The full K = 2–5 sensitivity sweep is therefore performed on the large-sample LESS simulations, whereas the size-limited field sampling plots used for LCC retrieval are partitioned at K = 3, which refines the two-component split into a purer high-reflectance component while keeping each component adequately sampled; the simulated VI–LCC comparison adopts the same K = 3 for consistency with the field data. The brightest component outperforms the all-pixel benchmark across the entire K = 2–5 range, so the conclusions do not hinge on a particular choice of K.

Importantly, this decomposition makes no explicit assumption about canopy structure: the GMM is fit to the 1-D distribution of apparent reflectance and requires no structural input (e.g., LAI, clumping index, or LAD) and no BRDF kernel. Its only implicit premise is distributional—that the within-canopy reflectance distribution contains a statistically separable high-reflectance subpopulation—so the approach is not tied to a specific canopy architecture; its applicability across canopy geometric types is examined in Section IV-D. Computationally, fitting the 1-D GMM by EM scales as $\mathcal{O}(NKI)$ for N vegetation pixels, $K \leq 5$ components, and I iterations, with each iteration linear in the number of pixels and the fit performed once as a preprocessing step. Empirically, on a single core of an AMD EPYC 9534 CPU, fitting and labeling the GMM (K = 3) on the 0.69 million vegetation pixels of a representative 3.6-mpixel scene took 1.2 s (1.9 s for K = 5), a single-core throughput of about 0.5 megapixels per second that scales near-linearly with pixel count (approximately 1.9 and 30 s for 1 and 10 megapixels, respectively). Because it solves no inverse problem, requires no ancillary data, and labels pixels independently, the workload parallelizes across image tiles, keeping the preprocessing negligible for the large data cubes typical of UAV hyperspectral monitoring.

Accordingly, we treat vegetation-pixel reflectance as a continuous distribution and use a K-component GMM to represent illumination-related subgroups for componentwise LCC retrieval. Each Gaussian component in the GMM represents a group of pixels with similar spectral reflectance features, which are typically observed under comparable local illumination conditions within the canopy. When the number of GMM components K = 2, the model generally distinguishes pixels into the predominantly sunlit portion and the predominantly shaded portion of the canopy. When the number of GMM components is increased to K = 3 (with K = 4 or 5 considered finer subdivisions thereof), the components can correspond to more detailed gradients of the light environment: the component with the lowest mean reflectance typically represents leaf areas in the lower to middle parts of the canopy that are deeply shaded by other foliage, where pixels primarily receive scattered light; the component with intermediate mean reflectance can represent leaves in the middle to upper parts of the canopy that are partially shaded, leaves in penumbral areas, or mixed pixels containing both sunlit and shaded leaf portions; and the component with the highest mean reflectance primarily represents leaves in the upper part of the canopy receiving more direct solar radiation. Typically (e.g., when the SZA is less than 60°), this latter component contains a larger proportion of leaves directly exposed to sunlight. The reflectance state of these leaves more closely reflects leaf-level optical responses under favorable illumination, and thus pixels in this component more effectively characterize leaf optical properties. Nevertheless, due to the 3-D structure and varying orientations of leaves, even directly illuminated leaves reflect different amounts of energy toward the sensor, resulting in a distribution of reflectance values within this component.

E. Pixel Clustering Based on the Existing Methods

To further evaluate the performance of GMM in extracting different illumination components within the canopy in this study, we compared its LCC estimation results with two other published pixel-selection baselines. Both baselines use hard thresholding (interval or percentile) on NIR apparent reflectance, rather than pixelwise posterior labeling. The first method draws inspiration from the approach of Wei et al. [27] for distinguishing sunlit and shaded portions from vegetation pixels. Specifically, after removing soil and other nonvegetation backgrounds, the reflectance values of the retained vegetation pixels in the 850-nm NIR band were extracted. A frequency histogram of these NIR reflectance values was constructed, and a bimodal Gaussian fitting was attempted. If the bimodal fitting was successful, pixels with NIR reflectance (ρ_{NIR}) satisfying $(\mu_1 + \sigma_1) < \rho_{\text{NIR}} < (\mu_2 + 2\sigma_2)$ were defined as "Gaussian sunlit pixels" using a histogram-statistics interval

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TABLE III

VIs SELECTED FROM THE LITERATURE. R_λ REPRESENTS THE REFLECTANCE AT BAND λ; R_{re} REPRESENTS THE REFLECTANCE AT THE INFLECTION POINT OF RED EDGE BAND, $R_{re} = (R_{670} + R_{780})/2$

VI names	Formulas	References
CI_{re}	$R_{800}/R_{730} - 1$	[55], [56]
mND_{blue}	$(R_{440} - R_{728})/(R_{440} + R_{850})$	[28]
MTCI	$(R_{754} - R_{709})/(R_{709} - R_{681})$	[51]
AIVI	$R_{735}/R_{720} - (R_{573} - R_{445})/(R_{573} + R_{445})$	[31]
CSI	$2.5 \times (R_{820} - R_{700})/(R_{820} + R_{700}) \times R_{450}/R_{700}$	[52]
REIP	$[(R_{re} - R_{700})/(R_{740} - R_{700})] \times 40 + 700$	[57]

extraction rule, based on the means (μ_1, μ_2) and standard deviations (σ_1, σ_2) of the two fit Gaussian distributions. If the bimodal Gaussian fitting was not applicable, a unimodal Gaussian fitting was adopted. In this case, pixels with NIR reflectance satisfying $\mu < \rho_{NIR} < (\mu + 2\sigma)$ were defined as "Gaussian sunlit pixels," based on the mean (μ) and standard deviation (σ) of the fit distribution. The second method partially draws on the pixel apparent reflectance selection idea from Jay et al. [4]. To ensure consistency with the GMM method of this study in the vegetation pixel extraction stage, vegetation pixels were first extracted using the NDVI threshold method detailed in the Supplementary Material. Then, for these extracted vegetation pixels, their reflectance values in the 850-nm NIR band were calculated and sorted in ascending order. The top 50% vegetation pixels with the highest NIR reflectance were selected as "percentile sunlit pixels," following the fixed percentile strategy in Jay et al. [4].

F. Estimation of LCC

1) Vegetation Index-Based Approach: Based on previous studies on retrieving LCC from canopy reflectance, we selected VIs defined within 400–900 nm from the literature. A simple ratio-based vegetation index, CI_{re}, was selected. The modified normalized difference blue index (mND_{blue}) was selected because it is sensitive to LCC and relatively insensitive to canopy structure [e.g., green fraction (GF) and green area index (GAI)]. As a ratio index, it is inherently robust to slope effects and illumination variations caused by slight cloud cover. mND_{blue} performs best when calculated from canopy components with higher within-component mean apparent reflectance and is effective at fine spatial resolutions (e.g., < 4.4 cm) typical of UAV data, aligning with this study's GMM approach for separating and using well-illuminated canopy components. Accurate radiometric calibration is important when using blue-band-based indices [28]. In addition, VIs based on the difference-division approach can often reduce the influence of soil and observation conditions, so MTCI was selected [51]. Furthermore, we included indices designed to reduce canopy-structural effects (e.g., LAI) on chlorophyll estimation, such as AIVI [31] and CSI [52]. The red edge region is the most sensitive region in the vegetation reflectance spectrum to changes in chlorophyll content, so the red edge inflection point (REIP) can accurately reflect subtle changes in vegetation chlorophyll content [53], [54]. The formulas for these VIs are shown in Table III.

A linear regression analysis was conducted between each VI and the corresponding LCC values, and the relationship was quantified using the coefficient of determination. This process was applied to both simulated and field datasets using 15 subsets of the reflectance spectra: 1) for the first subset, all vegetation pixels were used and 2) for the second to 15th subsets, component-specific pixels obtained from GMM separation were used (see Section II-C) with $K = 2, 3, 4$, and 5, resulting in 14 component subsets (K = 2, 3, 4, and 5 contribute 2, 3, 4, and 5 components, respectively, summing to 14). Physically, GMM aims to approximate the continuous distribution of canopy pixel reflectance.

Pixels were assigned to components based on the maximum posterior responsibility, and the reflectance spectra of all pixels within each subset were then averaged to derive VIs. The LCC estimation performance was evaluated for each of the 15 pixel subsets and six VIs for both simulated and field datasets. Then, the linear relationship between LCC and VI was determined based on the fitting results of the simulated dataset. Finally, the predictive performance of each relationship was quantified in terms of the coefficient of determination (R^2) and the rRMSE.

2) Radiative Transfer Model-Based Inversion: This study used an inverted PROSAIL model [58] to retrieve LCC. The PROSAIL model combines the leaf reflectance model (PROSPECT) [44], [45] with the canopy reflectance model (SAIL) [17], [59]. By inputting leaf biochemical parameters and canopy structural parameters, the PROSAIL model can simulate the canopy reflectance of a turbid-medium canopy at a certain viewing angle under specific illumination conditions. Although the turbid-medium assumption may not fully describe row-planted cotton canopies, PROSAIL inversion has been successfully applied for LCC retrieval in similar crops [60], [61].

Although pixel-level reflectance was used in the analysis, the selected pixels do not represent pure leaf-scale reflectance. Even at centimeter spatial resolution, a single pixel generally integrates contributions from multiple leaves with varying inclination angles, as well as within-canopy multiple scattering and residual background effects. Therefore, the reflectance of these pixels still corresponds to canopy-scale radiative behavior rather than the assumptions of a leaf-only model. In this study, PROSAIL was applied to a statistically constrained subset of canopy signals, rather than to canopy-averaged reflectance, allowing the model to be used under conditions that more closely satisfy its underlying assumptions. The reflectance spectra used as inversion input

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were GMM-component-averaged BRF: LESS-simulated BRF for the simulated dataset and UAV-measured hyperspectral reflectance for the field dataset.

We used the PROSPECT-D model [44] at the leaf scale to maintain consistency with the simulated dataset used in Section II-B. At the canopy scale, we employed the 4SAIL model [59], which is an optimized version of the SAIL model [17]. 4SAIL simulates bidirectional reflectance under a given solar-observation geometry defined by SZA, observer zenith angle, and relative azimuth angle (solar azimuth minus view azimuth). It uses LAI, leaf inclination distribution function, and hot spot parameter to describe the canopy structure, and dry/wet soil factor and soil brightness factor to account for soil spectral effects. Although the sensor is nominally nadir-viewing, an effective VZA of 4° was used to account for residual off-nadir viewing due to line-scanning geometry and platform attitude.

This study employs the differential evolution (DE) global optimization algorithm [62] to estimate LCC from reflectance data. The DE algorithm is an effective global optimization algorithm mainly used to solve optimization problems with real-valued parameters. Based on the idea of population evolution, the DE algorithm searches for the optimal solution through crossover, mutation, and selection operations during the iterative evolution process, effectively avoiding local optima. We used the DE function in scipy (version 1.12.0) [63] to implement the DE algorithm. To reduce the influence of factors other than LCC, the objective function was defined as the mean squared error (mse) between normalized reflectance spectra over 400–1000 nm, where each spectrum was normalized by its mean reflectance within this range. This normalization reduces sensitivity to multiplicative illumination scaling and emphasizes spectral shape cues relevant to chlorophyll absorption. In order to better conform to the actual situation of UAV agricultural monitoring, certain prior knowledge is set before inversion, including the range of parameters. The specific prior parameters are shown in Table IV. Unlike lookup table (LUT) approaches, this method calls PROSAIL as a forward model at each DE iteration; no precomputed LUT was generated.

The fixed (prior) and free parameters used in the inversion for both datasets are listed in Table IV. Specifically, several prior parameters were fixed to reduce model complexity. The soil reflectance parameters (dry/wet soil factor = 0.5 and soil brightness factor = 0.05) were set because soil pixels were largely removed by vegetation masking, and soil background influence was minimal under high vegetation cover. The soil brightness factor of 0.05 was further chosen to minimize any residual soil signal, while the dry/wet soil factor of 0.5 represents average moisture conditions. The leaf structure parameter (N) was fixed at 1.5, a typical mean value for healthy mesophytic leaves and consistent with the settings in our simulation dataset (see Table II). Brown pigment content (C_{brown}) was set to 0, as the study focused on the vigorous growing season when leaves were healthy and showed no senescence. Equivalent water thickness (C_w = 0.03 cm) and dry matter content (C_m = 0.01 g·cm^{-2}) were set based on field-measured averages for cotton leaves and were consistent with the simulation study. The hotspot parameter (hspot = 0.01) was set based on typical canopy structural conditions in the study area. These fixed parameters were intended to reduce the complexity of the inversion and ensure consistency with both the simulation part of this study and actual observation conditions. For the field dataset, PROSAIL inversion was applied only to the 2022 campaign. The 2023 field measurements were not included because substantial interannual differences in canopy structure and background conditions would violate the assumption of a consistent parameterization across campaigns. In contrast, the PLSR analysis was conducted using the combined field dataset to evaluate the robustness of the reflectance-component representation under a data-driven modeling framework.

TABLE IV
PRIOR KNOWLEDGE AND UNKNOWN PARAMETER SETTINGS FOR THE INVERSION PROSAIL MODEL

Parameter	Range / Values	Unit
Prior knowledge		
Dry/Wet soil factor	0.5	[−]
Soil brightness factor	0.05	[−]
Leaf structure parameter	1.5	[−]
Brown pigment	0	[−]
Equivalent water thickness	0.03	[cm]
Dry matter content	0.01	[g cm^{-2}]
Hot spot	0.01	[−]
Solar zenith angle	calculated by time and location	[°]
Observer zenith angle	4	[°]
Relative azimuth angle	calculated by time and location	[°]
Unknown parameters		
Leaf chlorophyll content	0–100	[μg/cm^2]
Carotenoid concentration	0–100	[μg/cm^2]
Leaf area index	0–10	[−]
Leaf angle distribution a	-1–1	[−]
Leaf angle distribution b	-1–1	[−]

Finally, for the simulated dataset, the model-input LCC was treated as the ground truth, whereas for the field dataset, SPAD-derived LCC (calibrated in Fig. 2) was used as the reference. Then, the prediction performance of each relationship was quantified based on the coefficient of determination (R^2) and the rRMSE.

3) PLSR-Based Estimation Using Measured Spectra: In addition to vegetation index-based regression and PROSAIL inversion, we further employed PLSR to examine whether the high-mean-reflectance component extracted by GMM yields consistent benefits within a multivariate full-spectrum modeling framework. This analysis was conducted exclusively on the field dataset. Prior to modeling, the measured reflectance spectra underwent Savitzky–Golay smoothing to suppress high-frequency noise while preserving spectral shape. Subsequently, the spectral range was restricted to 450–850 nm to focus on the chlorophyll-sensitive region, and the spectral predictors were mean-centered. PLSR models were constructed using two types of spectral inputs: 1) the mean spectrum of all vegetation pixels and 2) the mean spectrum of the high-mean-reflectance component (component 3) derived from the three-component GMM. Model evaluation employed a

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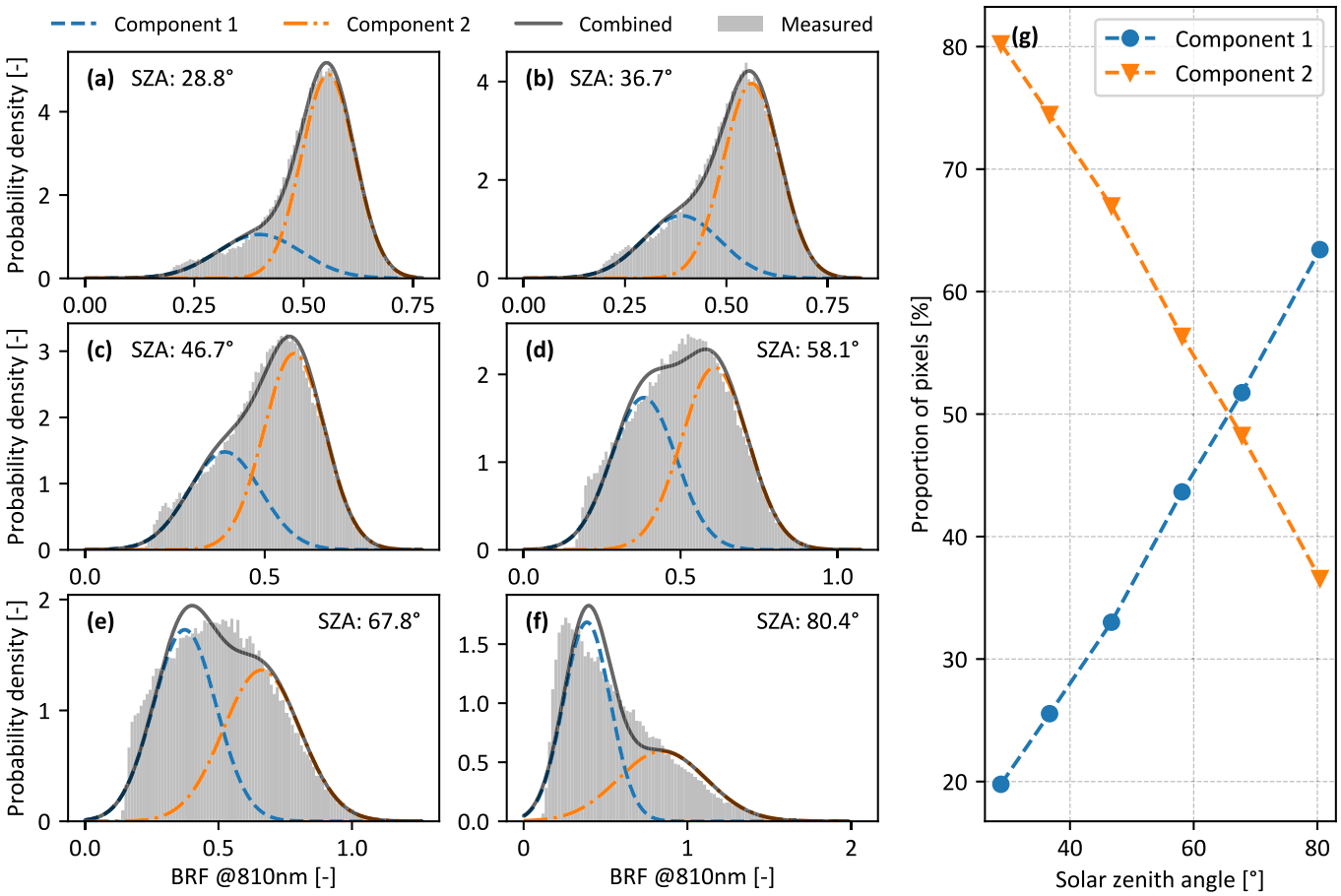


Fig. 4. Probability density functions (PDFs) of cotton canopy reflectance factor at 810 nm, modeled using a two-component GMM. (a)–(f) PDFs under varying SZAs. Component 1 (blue dashed) represents the lower-mean-reflectance canopy component (typically associated with shaded foliage), and Component 2 (orange dash-dot) represents the higher-mean-reflectance component (typically associated with sunlit foliage). The black solid line denotes the combined distribution, and the gray histogram denotes the measured PDF. (g) Pixel proportion of each component as a function of SZA.

double-split validation scheme with permutation resampling, and the number of latent variables was selected by minimizing the predicted residual error sum of squares (PRESSs) within the calibration set [64].

III. Results

A. Within-Canopy Reflectance Variability

Fig. 4 shows the probability density distribution of within-canopy reflectance at 810 nm under different SZAs for the hyperspectral data acquired on August 9, 2023. From Fig. 4(a)–(f), a two-component GMM provides an adequate approximation of the measured reflectance PDFs at relatively low to moderate SZAs (SZA = 28.8–46.7°). As SZA increases beyond this range, the measured PDFs progressively deviate from a simple bimodal form, showing increased skewness and broader tails, indicating that a two-component representation becomes less sufficient for fully characterizing the reflectance variability. Nevertheless, a two-component GMM is retained here as a standardized baseline to facilitate consistent quantification of component proportion changes with SZA and to enable comparison with more refined multicomponent representations introduced later. Fig. 4(g) shows that as the SZA increases from 28.8° to 80.4°, the proportion of component 1 increases from 20% to about 65% with a nearly linear trend, while the proportion of component 2 decreases synchronously. Unless the SZA is excessively high (greater than about 70°), the proportion of pixels occupied by component 2 remains larger. This suggests that canopy structural heterogeneity enhances mutual shading among leaves as SZA increases. Such shadows alter canopy-level scattering under different illumination angles, shifting the reflectance PDF and consequently changing the canopy-averaged reflectance. This conclusion is drawn under clear-sky conditions. However, under overcast conditions, although there are still high- and low-mean-reflectance leaf components in the canopy due to shading among leaves, the directional effect of illumination is no longer obvious due to the lower proportion of solar radiation. In addition, at high SZAs, the proportion of extremely high values (reflectance >1) at 810 nm increases noticeably, which may reflect geometry-driven anisotropy and, in field data, potentially enhanced specular contributions; previous studies also noted that specular reflection can become more significant at high SZAs [65], [66].

Fig. 4(g) illustrates the change in pixel proportions for each component as a function of the SZA when the GMM is set to two components (here, component 1 corresponds to the lower-mean-reflectance canopy component, typically

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associated with shaded foliage, whereas component 2 corresponds to the higher-mean-reflectance component, typically associated with sunlit foliage). In this study, the VZA of the UAV-borne sensor was close to 0° (i.e., nadir viewing). Under this observation geometry, as the SZA increases, the solar illumination becomes more oblique, significantly enhancing the mutual shading effect among leaves within the canopy. Consequently, the proportion of pixels in component 1 increases with increasing SZA. Conversely, the proportion of leaf areas directly receiving solar radiation decreases, leading to a synchronous decrease in the proportion of pixels in component 2, representing sunlit areas, as SZA increases. The GMM captures this shading-driven shift by quantifying how component proportions vary with the illumination geometry and canopy 3-D structure.

Because canopy structure modulates within-canopy shading, canopies with different structural heterogeneity are expected to exhibit distinct reflectance responses under varying illumination conditions, a conclusion that has also been reported at broader ecological scales [67]. To examine whether this structural–illumination coupling is observable at the fine spatial scale of UAV hyperspectral imagery in row crops, we measured canopy reflectance (400–1000 nm) from two adjacent cotton canopies (Canopy 1 and Canopy 2) under clear-sky conditions, with SZA ranging from 30° to 80°. Canopy 1 and Canopy 2 had similar leaf status (SPAD: 48–52), the same row planting pattern, and comparable LAI ($LAI_{canopy1} = 3.99$ and $LAI_{canopy2} = 3.61$); the two canopies were measured under the same sun–sensor geometry at each acquisition time. The plant height in both canopies was around 1.1 m, but Canopy 1 had a more uniform canopy height while Canopy 2 was more uneven. As shown in Fig. 5(b) and (c), the reflectance PDFs differed between the two canopies, likely due to their contrasting height uniformity and within-canopy occlusion. Here, a four-component GMM is used to resolve the structural contrast between the two canopies at a finer granularity than the two-component baseline, while keeping each component sufficiently populated given the per-canopy pixel count. At SZA = 67.8° and SAA = 272.2°, the reflectance PDFs at 810 nm for both canopies were unimodal, but their shapes differed [see Fig. 5(b) and (c)]. The probability distribution of Canopy 2 was clearly shifted to the left, meaning that the reflectance of Canopy 2 was lower. As shown in Fig. 5(d), the proportion of the lowest-mean-reflectance component increased with SZA for both canopies. However, because the plant height in Canopy 2 was not uniform, forming a canopy with varying heights, the proportion of leaves shaded by each other under oblique solar illumination was naturally higher. Therefore, the proportion of shaded leaf components in Canopy 2 was always higher than that in Canopy 1.

B. LESS-Based Performance of Different GMM Components on LCC Estimation

Fig. 6 shows the relationship between VI and LCC calculated using the component with the highest within-component mean reflectance (component 3) and the average reflectance of all vegetation pixels, after the pixel data of 5400 scenarios in the simulated dataset are divided into three components ($K = 3$) using a GMM. Here, it is pertinent to distinguish between the general term high-mean-reflectance components, which broadly refers to GMM-derived groups dominated by relatively well-illuminated pixels, and the highest-mean-reflectance component, defined as the component exhibiting the maximum within-component mean reflectance under a given GMM partition (K). In the three-component case ($K = 3$) shown here, this corresponds to component 3. Physically, the highest-mean-reflectance component is generally interpreted as representing leaf pixels experiencing relatively favorable illumination conditions with reduced structural shadow interference; consequently, its reflectance signal is considered to more closely approximate intrinsic leaf optical properties. This component is therefore selected as a refined subpopulation of pixels, expected to yield more robust LCC estimations due to its higher signal purity than mixed-pixel averages. The coefficient of determination R^2 varies across indices, with the highest R^2 achieved by AIVI when computed from component 3 (see Fig. 6). Across all investigated VIs, component 3 consistently outperforms the all-pixel case, with relative R^2 gains ranging from 5.3% [CSI: 0.75–0.79; Fig. 6(c1) and (c2)] to 33.3% [CI_{re}: 0.33–0.44; Fig. 6(b1) and (b2)]. This improvement supports the idea that selecting the highest-mean-reflectance component reduces within-canopy illumination mixing and provides a cleaner signal for LCC prediction than the all-pixel average.

An examination of Fig. 6 further reveals a consistent relationship between LCC and all investigated VIs for both simulated datasets (all pixels and component 3) and field-measured data, underscoring the general applicability of the proposed framework in capturing LCC–VI relationships.

Notably, AIVI, CSI, and MTCI show closer numerical agreement between the measured values (represented by orange hexagonal bins) and the simulated regression lines. This consistency can be related to their index design characteristics. In particular, CSI was proposed to be sensitive to chlorophyll while being relatively insensitive to LAI and other canopy or background factors (e.g., LAD and soil background), which helps reduce the influence of canopy-structural effects on chlorophyll retrieval [52]. Such properties contribute to more stable LCC–VI relationships when discrepancies exist between simulated and field canopy structures.

Conversely, other VIs such as CI_{re}, mND_{blue}, and REIP, while following the general trends, exhibit a more discernible offset in their absolute numerical values when comparing measured data to the simulated relationships. Several factors likely contribute to these discrepancies. First, real-world canopy structures are inherently more complex than their simulated counterparts, featuring variations in leaf clumping [18], LAD [20], and the presence of nonphotosynthetic canopy elements (e.g., stems and bolls) that are challenging to perfectly replicate in models.

Second, while radiometric calibration is applied, residual calibration errors and illumination variability can introduce systematic shifts. As highlighted by previous research [28], indices utilizing blue bands, such as mND_{blue}, are particularly sensitive to radiometric calibration and residual illumination effects and require meticulous radiometric calibration, where

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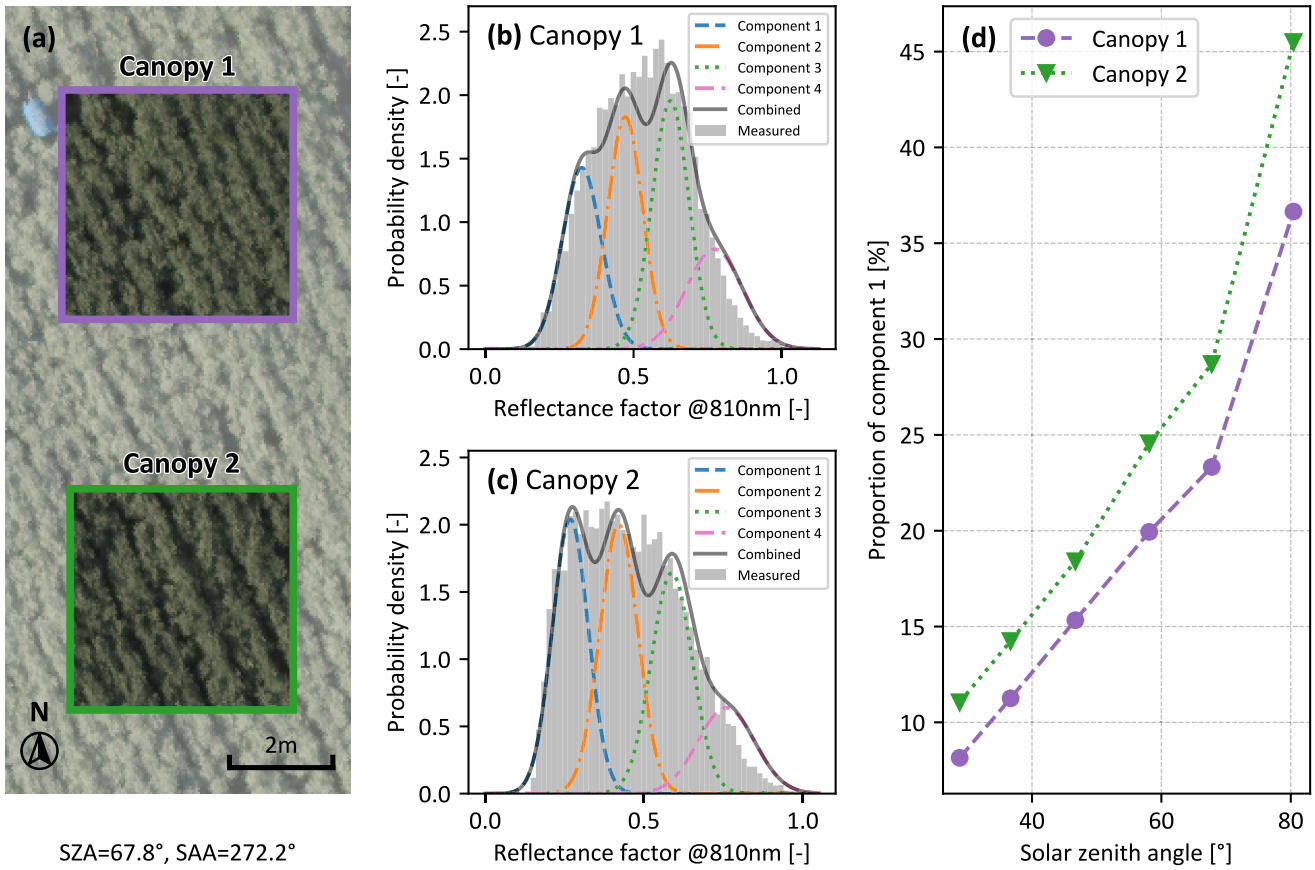


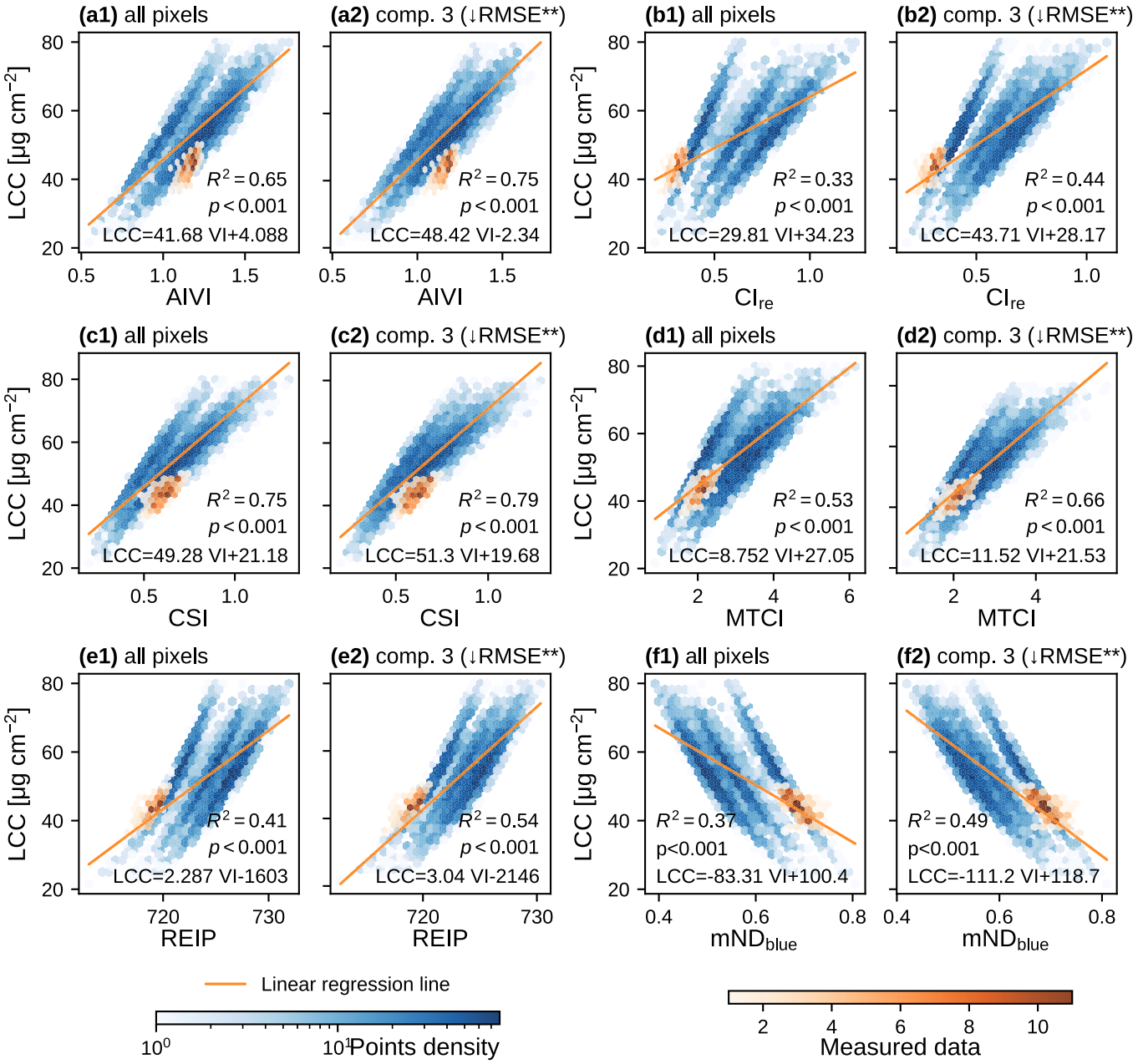
Fig. 5. Comparison of reflectance distributions for two adjacent cotton canopies with different height uniformity. (a) RGB views of Canopy 1 (more uniform height) and Canopy 2 (more heterogeneous height) under SZA = 67.8° and SAA = 272.2°. (b) and (c) PDFs of reflectance at 810 nm for (b) Canopy 1 and (c) Canopy 2, decomposed into four GMM components. The dark gray solid line denotes the combined distribution, and the gray histogram denotes the measured PDF. (d) Proportion of the lowest-mean-reflectance component as a function of SZA for both canopies.

any residual errors can lead to such shifts. Similarly, the precise optical properties of the soil background in the field may differ from those assumed in simulations, potentially affecting indices that are more sensitive to soil reflectance, even after vegetation masking.

Furthermore, simulations inherently rely on specific parameterizations for leaf optical properties (e.g., fixed ratios for carotenoids to chlorophyll [48], or fixed values for leaf water and dry matter content [49], as employed in the PROSPECT-D model setup for this study's simulations described in Section II-B). Natural variability in these leaf constituents in the field, beyond the distributions modeled, can lead to deviations in the spectral signatures and consequently in the VI values. For instance, the REIP, known for its sensitivity to chlorophyll content [53], [54], is highly sensitive to the exact shape of the red-edge slope, which can be influenced by a multitude of interacting biochemical and structural factors at the leaf level that may not be fully captured by the simulation parameters. Simple ratio indices like CI_{re} can also be more susceptible to additive atmospheric or background influences if not perfectly normalized. These factors, inherent in the comparison of simulated and empirical data, likely explain the observed differences in absolute VI values for some indices, despite the overall consistency in their response to LCC.

The results of LCC retrieval using PROSAIL inversion with componentwise mean reflectance from the LESS-simulated data are shown in Fig. 7. Fig. 7(a) shows that using the inverted PROSAIL model to retrieve LCC based on the average canopy reflectance may lead to underestimation of low values and overestimation of high values. From Fig. 7(b)–(f), it can be seen that the higher the mean apparent reflectance of the component (i.e., larger component ID in this ordering), the more accurate the LCC estimate based on its mean reflectance. R^2 gradually increases from 0.22 in Fig. 7(b) to 0.84 in Fig. 7(f). The rRMSE of component 5 is 39.8% lower than that of all pixels. The improved accuracy obtained using higher-mean-reflectance components compared with the all-pixel average suggests that these components are less affected by illumination variability and canopy structural mixing. Consistently, components with higher mean reflectance show smaller rRMSE, indicating more accurate LCC retrieval. Low-mean-reflectance components, dominated by scattered and transmitted light, can alter spectral shape and normalization, and may therefore bias LCC retrieval high in some cases. Overall, analyzing the reflectance distribution and separating higher-mean-reflectance vegetation components via GMM improves LCC retrieval with PROSAIL inversion compared with using the all-pixel average.

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** indicates a statistically significant reduction in RMSE compared with using all pixels (p < 0.01).

Fig. 6. Linear relationships between LCC and VIs for simulated data using all vegetation pixels and the highest-mean-reflectance component under $K = 3$ (component 3). Component 3 corresponds to the GMM component with the highest within-component mean apparent reflectance and is mainly composed of relatively well-illuminated (sunlit-dominated) leaf pixels. (a1)–(f1) VI–LCC relationships using all pixels. (a2)–(f2) VI–LCC relationships using only component 3 pixels. The orange line denotes the fitted linear regression; the regression equation, R^2, and p value are reported in each panel. Simulated samples are displayed as blue hexagonal bins with density indicated by the left color bar, and field-measured data are overlaid as orange hexagonal bins with density indicated by the right color bar for comparison.

1) Comparison of LCC Estimation Performance Across GMM Components and Existing Methods: Fig. 8 compares the R^2 between VIs and LCC across GMM partitions with $K = 2, 3, 4$, and 5, using the all-pixel result as a reference. Overall, R^2 tends to increase when VIs are computed from components with higher within-component mean reflectance (i.e., larger component IDs), and the highest-mean-reflectance component consistently outperforms the all-pixel benchmark, whereas lower-mean-reflectance components yield lower R^2. For indices such as CI_{re}, mND_{blue}, MTCI, AIVI, CSI, and REIP, the brightest component under $K = 5$ provides higher R^2 than using all pixels together, indicating that separating high-mean-reflectance vegetation components via GMM can improve LCC prediction.

To further evaluate the GMM method, we compared it with two existing methods based on simulation data: a histogram-based Gaussian-fit threshold on NIR reflectance ("Gaussian sunlit," following Wei et al. [27]), and a fixed apparent

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reflectance percentile selection on NIR reflectance after NDVI-based vegetation masking ("percentile sunlit," inspired by Jay et al. [4]). As shown in Fig. 8, the highest-mean-reflectance components obtained from multicomponent partitioning (K = 3–5) generally yield higher R^2 than the two baselines, whose R^2 values are approximately 0.38–0.76 ("Gaussian sunlit") and 0.40–0.78 ("percentile sunlit"), depending on the VI. These two baselines span the published approaches for illumination-based pixel selection in single-nadir UAV hyperspectral retrieval: one adapts to each scene's NIR reflectance histogram; the other applies a fixed reflectance percentile. Notably, even under the bi-component setting (K = 2), the higher-mean-reflectance GMM component generally outperforms the histogram-thresholding baseline ("Gaussian sunlit"), while showing comparable performance to the percentile-based baseline ("percentile sunlit") for several VIs. Overall, increasing K from 2 to 3–5 enables GMM to more consistently isolate an optimal high-mean-reflectance component with higher R^2 than both the two baselines and the K = 2 partitioning. We further justified the choice of K with a sensitivity analysis based on Fig. 8. For all six VIs the R^2 of the brightest component increased monotonically with K but with diminishing increments—the gain from K = 4 to K = 5 was roughly half that from K = 2 to K = 3 (for CI_{re}, successive gains of 0.045, 0.025, and 0.020 across $K = 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$)—while the brightest-component mean reflectance rose only modestly (from 0.30 at K = 2 to 0.35 at K = 5). The five components are, moreover, not interchangeable: at K = 5 the per-component mean reflectance rises monotonically from 0.17 (darkest) to 0.35 (brightest) and the retrieval R^2 rises in step (e.g., CI_{re} from 0.18 to 0.49, and the PROSAIL-inverted R^2 from 0.22 to 0.84; Figs. 7 and 8), so each component carries a distinct mixture of illumination and structure and the brightest one retains the least structurally contaminated signal. Physically, a finer partition isolates a smaller, brighter component increasingly dominated by direct or low-order scattering from well-illuminated leaf surfaces; the saturating gains, together with the shrinking pixel population and the risk that an over-refined brightest component becomes dominated by geometric extremes (hotspot or specular pixels with apparent reflectance >1), motivate capping K at 5 as a balance between component purity and interpretability.

C. Evaluations Based on Field Measurements

After classifying cotton canopy vegetation pixels into three illumination-related components using a GMM on the measured data, we calculated VIs using the mean reflectance of component 3 (the component with the highest within-component mean reflectance, predominantly associated with sunlit leaves) and the mean reflectance of all vegetation pixels. Fig. 9 shows the results calculated from hyperspectral data acquired at different times on August 9, 2023. Dozens of cotton canopies with different canopy structures were investigated, with LAI ranging from 2.31 to 5.66. Overall, compared with VIs calculated from all vegetation pixels, those calculated from the high-mean-reflectance component (component 3) are more concentrated across different LAI (canopy-structure) conditions, indicating a reduced influence of LAI on VI. The standard deviation of the VIs calculated from the high-mean-reflectance component was generally reduced by more than 50% compared to those calculated from all vegetation pixels, indicating improved stability across different LAI values. For the same VI, component-3 curves were often more stable than the all-pixel curves, as indicated by smaller absolute slopes in the VI–SZA regressions (CI_{re} and REIP excluded).

The correlations between LCC and VIs calculated using the mean reflectance of the high-mean-reflectance component (component 3) are shown in Fig. 10. Among all VIs, the high-mean-reflectance component generally exhibits better correlations than using all vegetation pixels, and its point distribution is more concentrated. The increase in R^2 reaches up to 31% [see Fig. 10(f1) and (f2)], indicating that the high-mean-reflectance component is less sensitive to illumination variability and canopy-structure effects, consistent with the simulated results. After grouping pixels based on the probability distribution of within-canopy reflectance, the VI–LCC relationships established using component 3 are the most robust within this analysis. This pattern holds for all VIs tested in the field dataset.

The PROSAIL inversion results in Fig. 11 show that LCC estimates are generally more concentrated for the high-mean-reflectance component (component 3) than for all vegetation pixels. The rRMSE is reduced by 12.9%, and the coefficient of determination is increased by 39.5%. This pattern is consistent with the simulated data, indicating that PROSAIL-based LCC inversion benefits from first using a GMM to separate high-reflectance vegetation pixels based on the probability distribution of canopy reflectance.

For the measured data, PLSR was further employed to evaluate whether the advantages of the spectral components extracted by GMM are independent of the specific retrieval model. When the mean spectrum of all vegetation pixels was used as input, the PLSR estimation accuracy for LCC was moderate (mean validation R^2 = 0.46, RMSE = 2.23 $\mu g \cdot cm^{-2}$). In contrast, based on the spectral input of the highest-mean-reflectance component (component 3), the PLSR results exhibited a higher R^2 and lower error (mean validation R^2 = 0.56, RMSE = 2.03 $\mu g \cdot cm^{-2}$; Fig. 12). This improvement was statistically significant across the repeated double-split permutations (paired Wilcoxon signed-rank test, $p < 0.001$ for both R^2 and RMSE). The magnitude of this improvement is consistent with the results observed in VI regression and PROSAIL inversion.

To assess whether this improvement is robust to data partitioning and model-complexity selection, we further conducted a repeated double-split validation with permutation resampling and PRESS-based component selection for both spectral inputs, as summarized in Fig. 13.

Taken together, the consistent improvements observed for VI-based regression, physics-based PROSAIL inversion, and data-driven PLSR indicate that the benefit of the GMM-derived reflectance components arises at the level of reflectance representation, rather than from any specific retrieval methodology.

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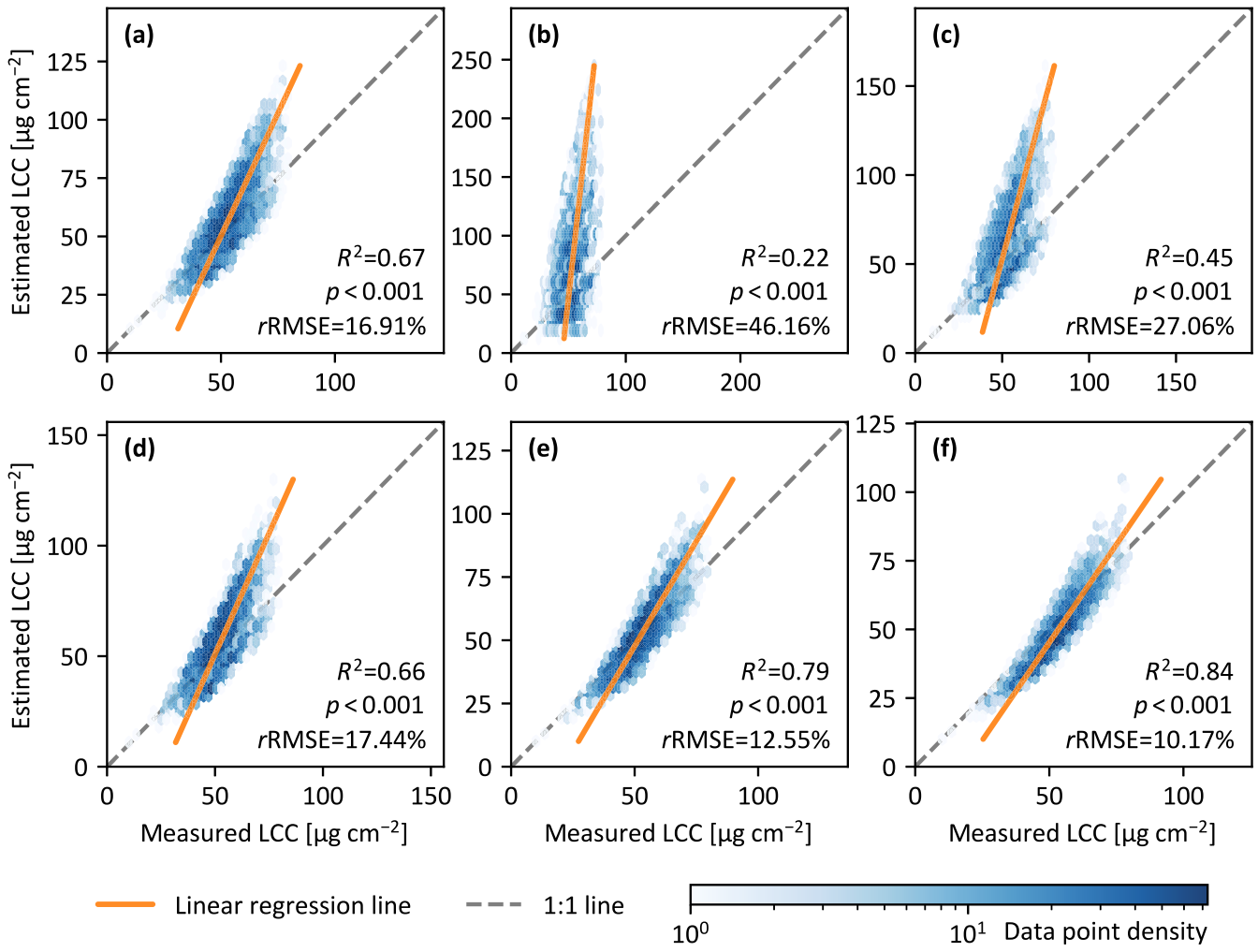


Fig. 7. Comparison of retrieved LCC from PROSAIL inversion against model-input LCC across different canopy components. These components were identified by classifying pixels using a GMM with five components, ordered by their mean reflectance from the lowest (b) to the highest (f). (a) Aggregated results for all pixels. (b)–(f) Results for individual components derived from the GMM. The orange solid line and the black dashed line represent the linear regression fit and the 1:1 line, respectively. Point density is represented by a color gradient from white to blue. The coefficient of determination (R^2), the rRMSE, and the p value are shown.

IV. Discussion

A. Resolving the Within-Canopy Reflectance Variability

Accurate retrieval of LCC from UAV hyperspectral imagery critically depends on how within-canopy illumination heterogeneity is represented. Variations in solar geometry and canopy structure generate continuous reflectance gradients across vegetation pixels, which can substantially influence canopy-scale LCC estimation. To mitigate illumination effects, many remote sensing studies partition vegetation canopies into sunlit and shaded components and analyze them separately [5], [68], [69], [70], [71]. This sunlit–shaded separation offers a practical but simplified representation of canopy illumination conditions; however, centimeter-level UAV imagery often exhibits a more continuous, multilevel reflectance variability that is not fully captured by a binary partitioning.

From a radiative transfer perspective, the advantage of the high-mean-reflectance component comes from retaining pixels whose signals are dominated by direct illumination and low-order scattering from leaf surfaces. These pixels are not pure leaf reflectance, but their chlorophyll-absorption features are less diluted by shadowing, canopy structure, and multiple scattering than spectra averaged over all vegetation pixels. This is a relative purification rather than a recovery of pure leaf reflectance, and the contamination is band-dependent. In the NIR, where leaf reflectance and transmittance are high, canopy reflectance is more strongly affected by multiple scattering and therefore shows large illumination-driven brightness variability; this makes the NIR sensitive for *separating* illumination components, even though it carries little chlorophyll signal. The LCC signal instead resides in the chlorophyll-sensitive visible-to-red-edge region (the influence of LCC on canopy reflectance is concentrated below ~ 800 nm), where the contrast between low red reflectance and the red-edge rise makes changes in direct illumination, shadow fraction, and local leaf orientation strongly alter the apparent absorption depth. No canopy pixel is free of multiple scattering and structural mixing, so brightness is a necessary but not sufficient indicator

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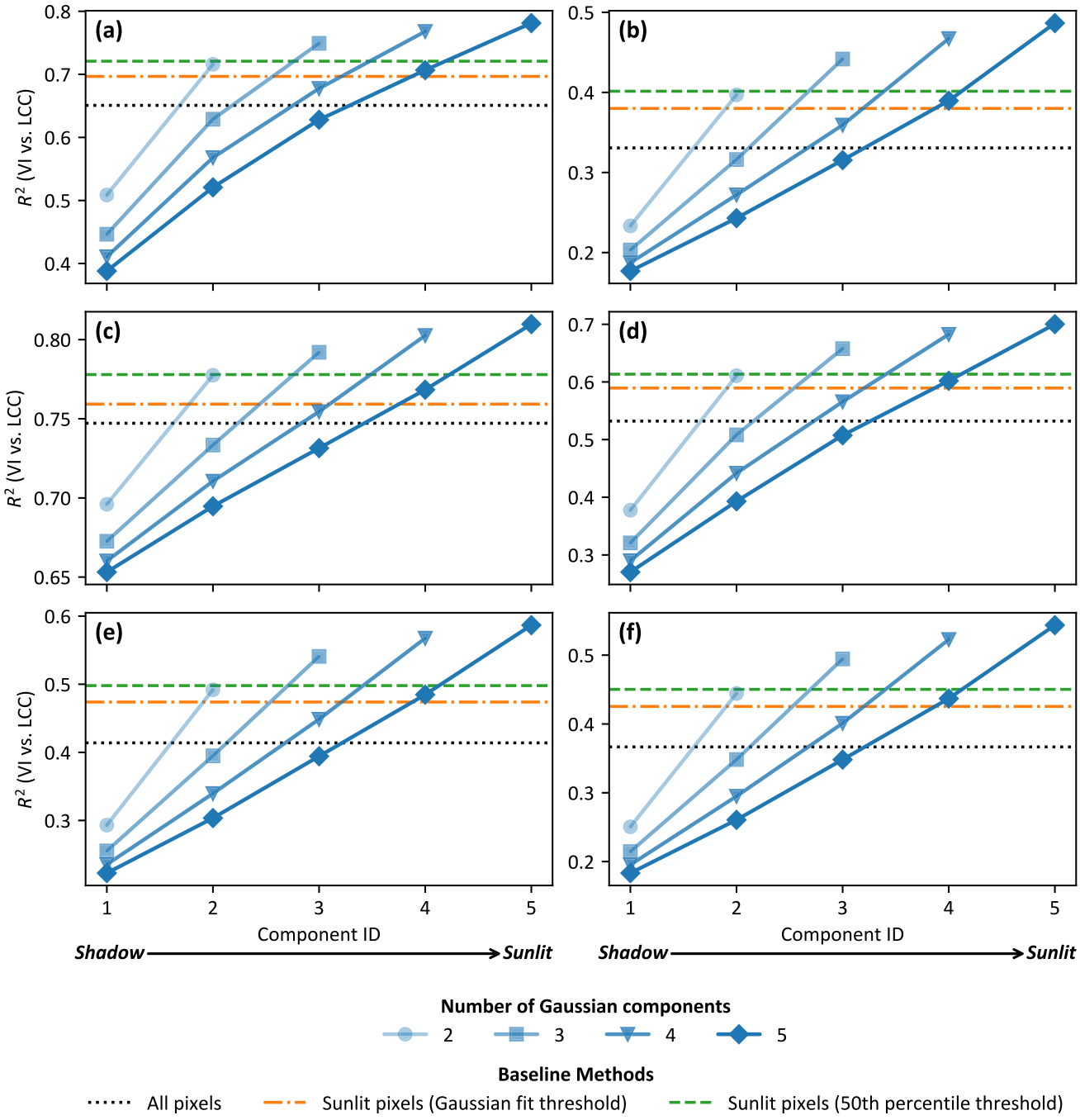


Fig. 8. Relationship between LCC and VIs across GMM components. Vegetation-pixel reflectance distributions were modeled using an n-component GMM with n = 2–5. (a)–(f) Coefficient of determination (R^2) between LCC and each VI for every GMM component. Components are indexed from 1 to n and ordered by mean apparent reflectance, where component 1 has the lowest mean (more shadow-dominated) and component n has the highest mean (more sunlit-dominated). The black dotted line denotes the R^2 obtained using all vegetation pixels without GMM separation. The orange dash-dotted line and green dashed line denote the R^2 obtained using the threshold-based "Gaussian sunlit" pixels (following Wei et al. [27]) and the "percentile sunlit" pixels (50th percentile; following Jay et al. [4]), respectively. (a) AIVI. (b) CI_{re}. (c) CSI. (d) MTCI. (e) REIP. (f) mND_{blue}.

of illumination state. The framework therefore does not invert ρ_L, which is ill-posed for single-view, centimeter-scale pixels; it selects the observed pixels whose spectra are least contaminated by illumination heterogeneity. Because brightness is an observable proxy for this illumination state, the benefit of selecting the highest-mean component derives from retaining well-illuminated, low-shadow pixels; the probabilistic framework contributes the scene-adaptive, preset-free placement of this selection rather than a transformed spectral signal.

This selection-based perspective also distinguishes our framework from illumination- and shadow-correction strategies that transform reflectance, such as bidirectional-reflectance normalization [21], [22], [23] and sunlit–shaded correction [24], [26]. Because the leaf signal and canopy

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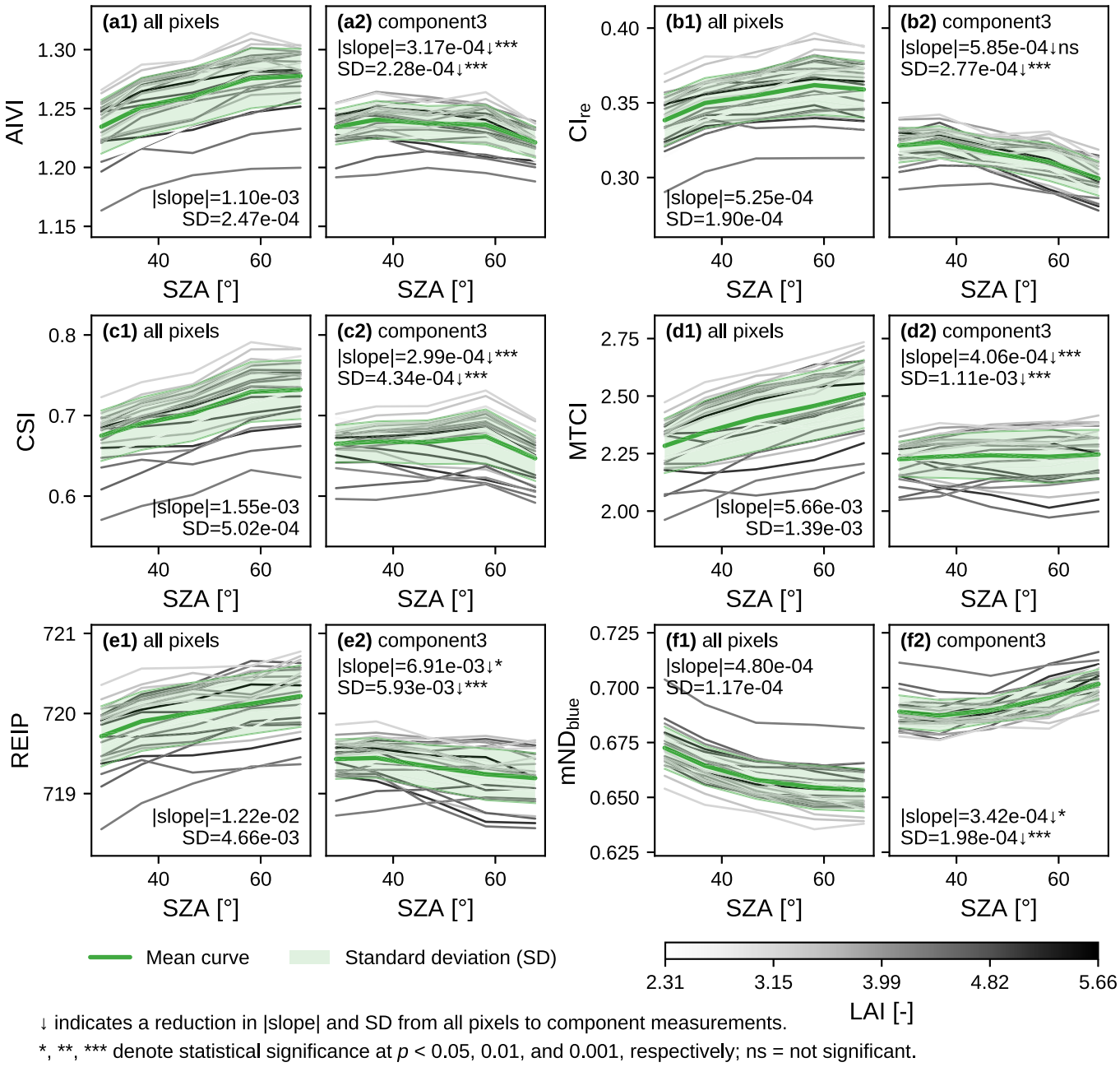


Fig. 9. VIs calculated using the mean reflectance of the high-mean-reflectance component (component 3) and all vegetation pixels for cotton canopies with varying LAI on August 9, 2023. The VIs were calculated under five SZAs. Panels (a1) and (a2) show AIVI; (b1) and (b2) CIre; (c1) and (c2) CSI; (d1) and (d2) MTCI; (e1) and (e2) REIP; and (f1) and (f2) mNDblue. In each pair, the panel ending in '1' shows results calculated using all vegetation pixels, whereas the panel ending in '2' shows results calculated using component 3. The green line represents the mean VI across all canopies, the green shaded area indicates ±1 standard deviation of VI values across canopies with different LAI at each SZA, and the gray lines denote the field measurements, with darker tones indicating higher LAI.

structure are entangled multiplicatively and nonlinearly in ω_{canopy}, such a transform must assume a structure or kernel; when that assumption departs from reality, the residual error is wavelength-correlated and can mimic a biochemical signal, biasing the retrieval. A correction can at best preserve, and generally loses, the leaf information already mixed by the canopy; it cannot restore information that the observation no longer contains. Selection instead keeps the least-contaminated pixels unaltered, introducing no structural assumption and propagating no kernel uncertainty; its only cost, a reduced spatial coverage, is transparent, whereas a transform reshapes the very signal used for retrieval.

The widespread application of UAV remote sensing in agriculture and environmental monitoring has made high-resolution data analysis routine but has also introduced new challenges. A central issue is how to extract, from canopy-scale reflectance, the spectral signals that best represent leaf optical properties and link to leaf functional traits such as LCC. This challenge is often described as a scale mismatch or a representative-pixel problem. From a radiative transfer

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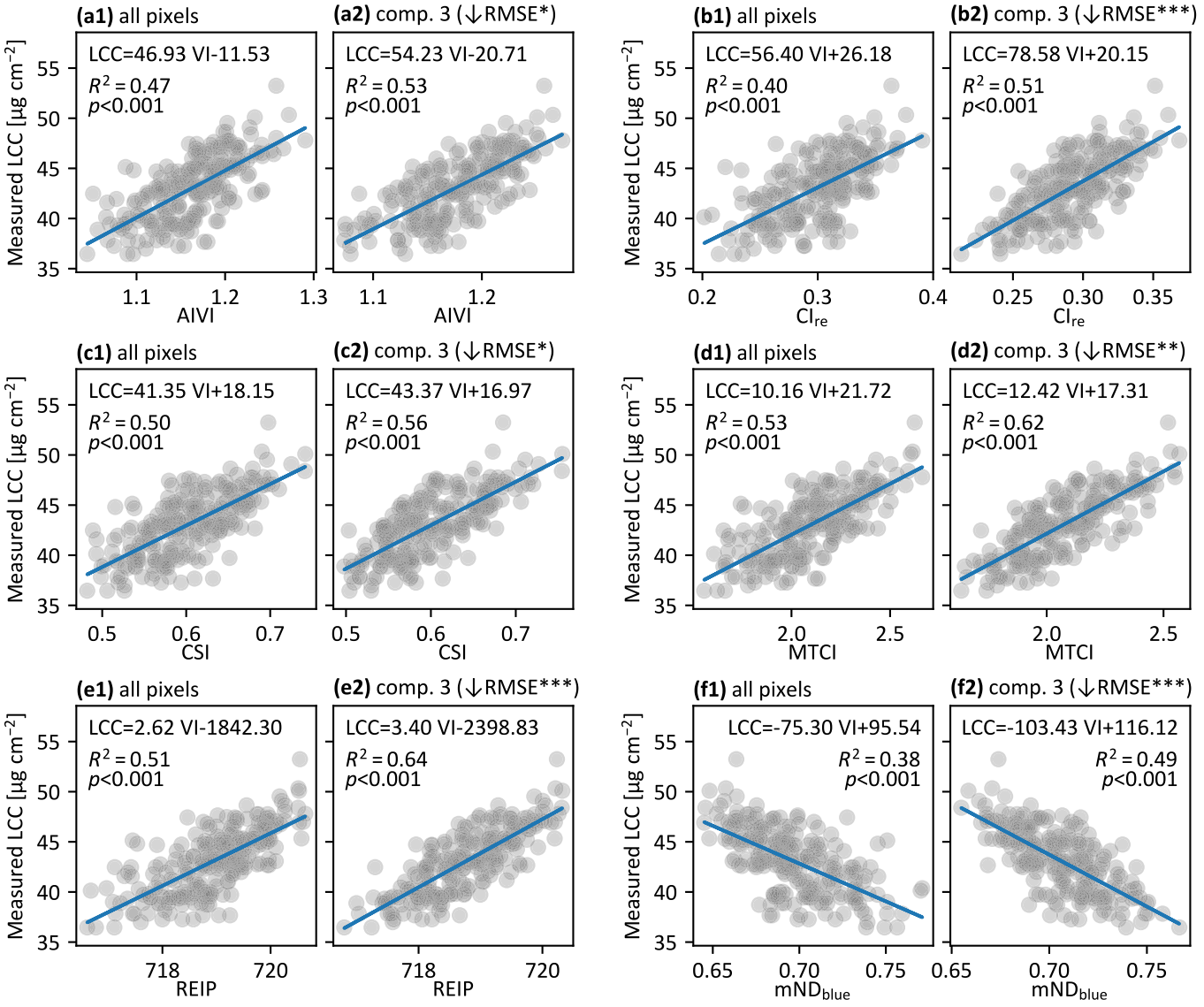


Fig. 10. Linear relationships between LCC and VIs for the field-measured data. Component 3 (comp. 3) corresponds to the highest within-canopy mean apparent reflectance (mostly sunlit leaves) identified using a three-component GMM. (a1)–(f1) VIs calculated using all vegetation pixels. (a2)–(f2) VIs calculated using only component 3 pixels. The blue line denotes the regression line, and the gray points represent the field measurements.

perspective, dividing vegetation pixels into sunlit and shaded parts is a practical data-analysis method [4], [5], [70]. However, this dichotomy assumes an idealized situation in which irradiance is uniform within each sunlit or shaded class, which is not the case in real canopies. Here, the main advance is pixel-level component assignment that preserves within-canopy heterogeneity. A key finding (see Fig. 8) is that the multicomponent GMM ($K > 2$) yields higher LCC-estimation accuracy than both the bi-component GMM ($K = 2$) and the two published bi-component pixel-extraction methods. Specifically, with $K = 2$, the higher-mean-reflectance cluster still spans a relatively broad range of illumination conditions, whereas a finer partition ($K = 3$–5) isolates a smaller, brighter, lower-shadow component, which improves retrieval accuracy. When the GMM uses more clusters ($K = 3$–5), the highest-mean-reflectance cluster achieves higher R^2 values than both the bi-component GMM and the Gaussian sunlit and percentile sunlit methods. These results suggest that, for high-resolution UAV hyperspectral imagery, partitioning pixels into only two illumination classes is often insufficient for accurate LCC retrieval compared with a multicomponent strategy. In reality, due to random leaf inclination and mixed sources of irradiance, the irradiance received by leaves varies continuously within the canopy, and the reflectance of pixels in top-of-canopy hyperspectral imagery also varies continuously. Because the TOC reflectance distribution is shaped by canopy structure, it can also provide information on canopy structural heterogeneity [72]. These results indicate

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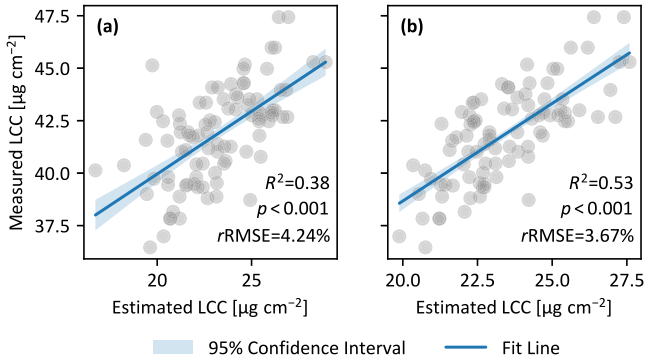


Fig. 11. Comparison between measured LCC and LCC estimated from PROSAIL inversion for (a) all vegetation pixels and (b) high-mean-reflectance component (component 3) identified by the GMM. The blue solid line denotes the linear fit, and the gray points denote the field measurements.

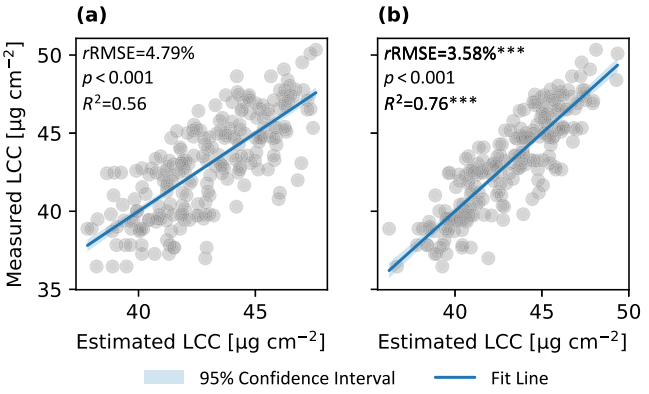


Fig. 12. Comparison between measured LCC and LCC estimated from PLSR for (a) all vegetation pixels and (b) component 3 (high-mean-reflectance component) identified by the GMM. The blue solid line denotes the linear fit, and the gray points denote the field measurements.

that a finer, componentwise representation can better capture within-canopy illumination heterogeneity and improve the stability of vegetation-parameter retrieval. This convergence across retrieval approaches reflects that isolating illumination-consistent reflectance components improves the physical representativeness of the spectral signal, thereby reducing the confounding effects of canopy structure and illumination irrespective of the specific modeling strategy. Increasing the number of components is then a natural extension, especially under stronger heterogeneity. This componentwise view also provides a practical link between reflectance distributions and canopy structural heterogeneity.

In practice, since low-altitude and ground-based data platforms can collect data under cloudy and overcast conditions [6], and irradiance under these natural conditions is affected by meteorological factors, dividing the canopy into sunlit and shaded components may have significantly different effects in overcast conditions compared to sunny conditions. This is because the direct irradiance component can be strongly reduced under overcast conditions. However, mutual shading among canopy branches and leaves and multiple scattering within the canopy still occur on overcast days, and leaf components with different apparent reflectance levels still exist. In this case, it is more practical to analyze vegetation canopy information from the continuous distribution of vegetation pixels based on the GMM method. This point is also supported by Wei et al. [27], who demonstrated that under diffuse illumination conditions, modeling the reflectance distribution of vegetation pixels using Gaussian-based approaches can effectively improve LCC estimation by reducing the confounding effects of heterogeneous illumination.

On sunny days, many crops, including rice, wheat, and corn, can exhibit specular reflection [73]. Under a near-nadir UAV viewing geometry, this phenomenon is particularly evident in near-noon acquisitions with small SZA and may lead to abnormally high apparent reflectance for a subset of vegetation pixels. Specular reflection mainly occurs at the air-epidermis interface [74] and contributes limited vegetation physiological information [48]. Therefore, from an acquisition-planning perspective, conducting UAV observations at moderately higher SZAs is more likely to reduce specular contamination compared to near-noon acquisitions, and the higher SZA also increases the canopy ground-projected shadows, which can reduce the contribution of soil background to the measured signal [32]. Consequently, analyzing the continuous distribution of vegetation pixels using a GMM framework provides a more robust way to isolate representative canopy components across varying illumination conditions. In addition, when vegetation is under stress, canopy structural heterogeneity is often higher, forming a distinct high-low staggered structure in which shorter plants are easily shaded by taller plants. A simple dichotomy tends to overlook the information from these shorter or partially shaded plants and may lose stress-related spectral signals, whereas a distribution-based analysis can better preserve such heterogeneity information.

B. Generalizability: Canopy Spatial Aggregation Governs the Benefit

A central question is whether the benefit of selecting the high-mean-reflectance component is specific to the cotton canopies studied here or follows a generalizable structural principle. To isolate the governing variable, we define the illumination-variance suppression fraction $f_s = (\sigma^2_{\text{all}} - \sigma^2_{\text{bright}})/\sigma^2_{\text{all}}$, the fractional reduction in within-LCC variance (averaged across SZA) obtained by restricting the vegetation index to the brightest GMM component rather than using all vegetation pixels; positive f_s indicates that bright-component selection reduces SZA-induced scatter, and negative f_s the opposite.

Using the LESS radiative transfer model, we varied only the canopy clumping index Ω while holding all other canopy and illumination parameters fixed. The CI_{re} f_s decreases monotonically from +0.89 at $\Omega = 0.55$ (spatially aggregated canopy) to −1.09 at $\Omega = 1.00$ (random distribution), with MTCI following the same trend [see Fig. 14(a) and Supplementary Table S2], identifying Ω as the direct causal variable and locating the sign change near $\Omega \approx 0.75$ as the applicability boundary. The underlying mechanism is that spatial aggregation—whether from row planting or discrete crown spacing—organizes coherent sunlit cluster tops whose brightest pixels are better illuminated and less affected by

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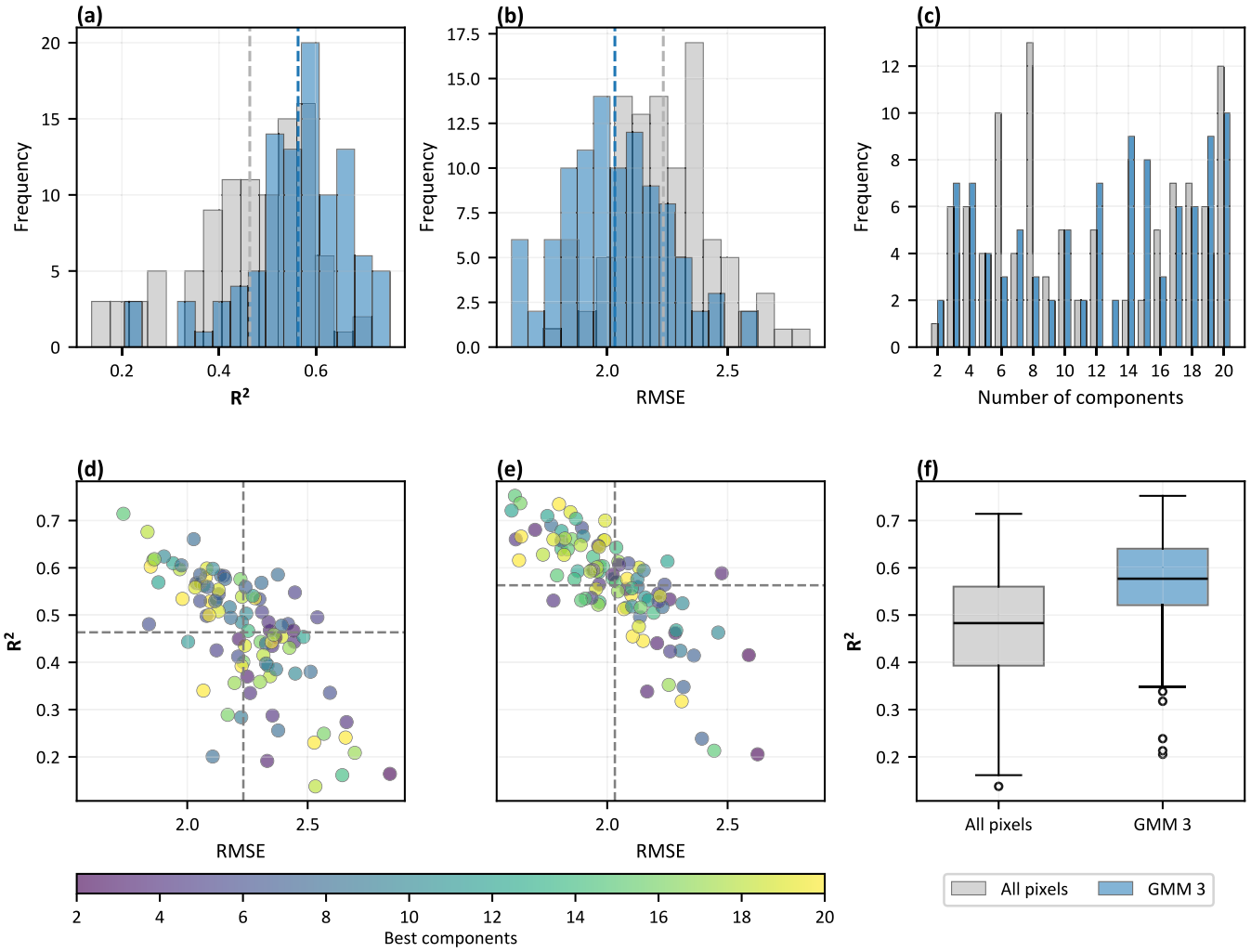


Fig. 13. Robustness assessment of PLSR models via double-split validation, comparing spectral inputs from all vegetation pixels (gray) and the highest-mean-reflectance component (GMM3, blue). Distributions of validation (a) R^2 and (b) RMSE, with dashed lines indicating means. (c) Frequency of optimal latent variables. (d) and (e) Scatterplots of R^2 versus RMSE colored by model complexity (number of latent variables). (f) Boxplot summary of R^2 distributions.

shadowing; in structurally uniform, closed canopies ($\Omega \to 1$) this organization is absent and the approach does not apply.

To test generality across vegetation types, we simulated four structurally diverse canopies under identical illumination: row-sown cotton, row-sown soybean, and two RAMI-benchmark deciduous tree species—silver birch (*Betula pendula*) and European ash (*Fraxinus excelsior*)—at 4-m crown spacing. All four yield positive CI_{re} f_s [+0.37 to +0.90; Fig. 14(b) and Supplementary Table S2]. The European ash result is particularly informative: despite a near-erectophile LAD (area-weighted $\cos\theta = 0.34$), its isolated crowns still organize coherent sunlit cluster tops that dominate component selection, yielding $f_s = +0.88$ comparable to the planophile cotton canopy. Canopy spatial aggregation at the crown or row scale—not leaf angle per se—therefore governs the method's effectiveness. The observational data already presented corroborate this at no additional measurement cost: the two adjacent cotton canopies in Fig. 5 that differ in canopy-top height uniformity display distinctly different BRF multimodality, precisely the Ω-dependent separability described above.

C. Effect of Solar Angle on Within-Canopy Reflectance Variability

Changes in SZA not only rescale canopy reflectance but also reshape the distribution of within-canopy reflectance at centimeter resolution. In Fig. 3, we observe a systematic redistribution of TOC reflectance with increasing SZA, consistent with angle-dependent canopy anisotropy arising from illumination–view geometry and canopy structure [13], [14], [16], [75]. With increasing SZA, the distribution shifts toward lower values while the upper tail becomes heavier, so the mean does not necessarily decrease; in our observations, SZA >67.8° yielded a higher mean reflectance than SZA = 28.8° in the shown band. The GMM-based analysis suggests that high-reflectance pixels become more frequent as SZA increases. LESS simulations indicate that this pattern is not solely driven by specular reflection because the simulated leaves are Lambertian. Two mechanisms can explain the heavier upper tail. First, at large SZAs, a subset of leaf facets can transmit a larger fraction of direct sunlight toward the nadir sensor along favorable geometric paths, producing unusually bright pixels

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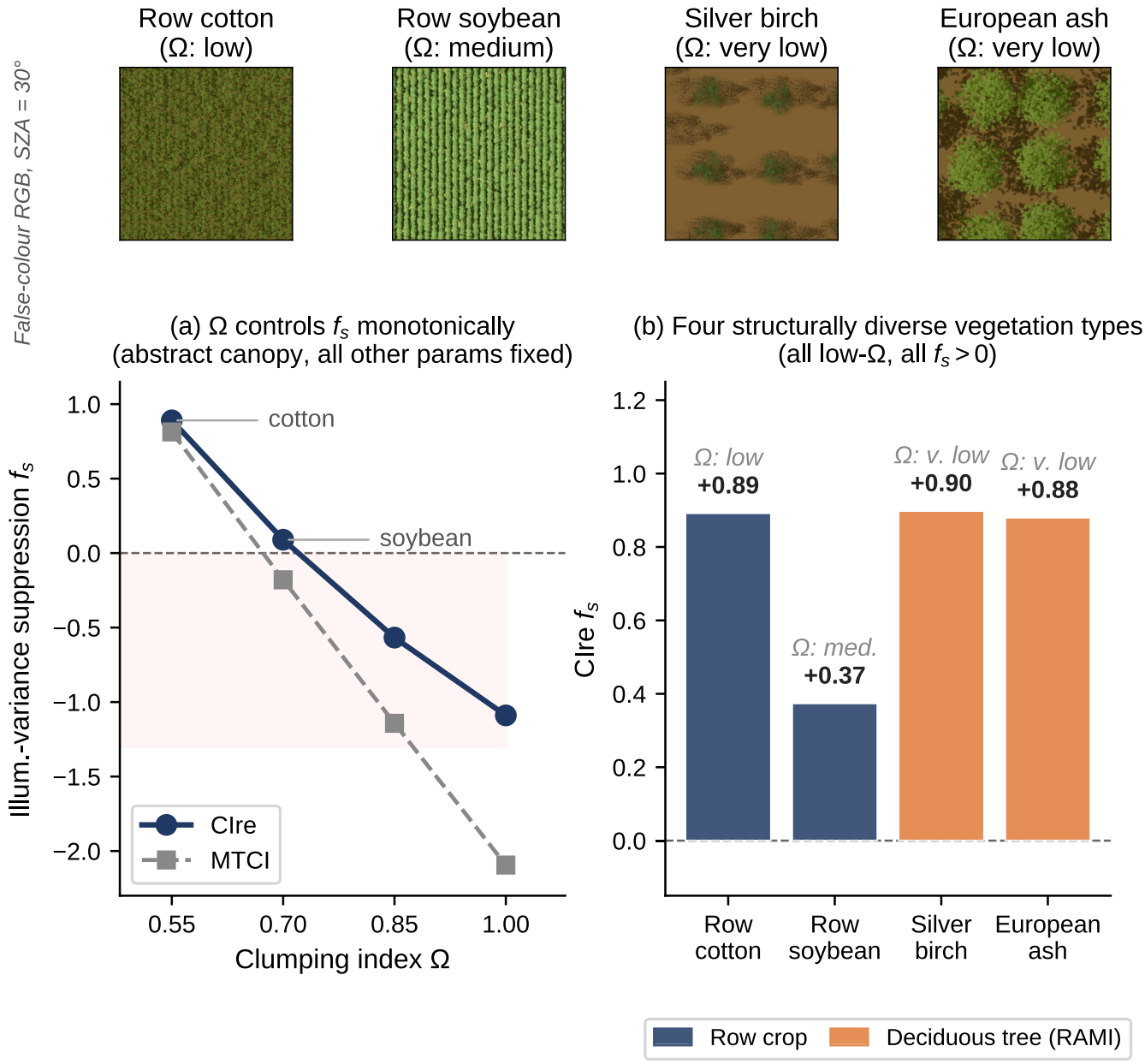


Fig. 14. Causal role of canopy clumping index Ω in governing the CI_{re} illumination-variance suppression fraction f_s. The top row shows false-color RGB renderings (CIE 1931 standard observer, D65 illuminant; SZA = 30°) of each simulated canopy type, illustrating the structural contrast between spatially aggregated (row crops, isolated crowns) and uniform canopies. (a) Controlled LESS simulation in which Ω alone was varied while all other parameters were held fixed: f_s decreases monotonically from + 0.89 at Ω = 0.55 (spatially aggregated) to −1.09 at Ω = 1.00 (random distribution), with MTCI following the same trend. The dashed line marks $f_s = 0$ (no-benefit boundary); the red-shaded region indicates where bright-component selection amplifies rather than suppresses SZA-induced scatter. (b) CI_{re} f_s for four structurally diverse vegetation types simulated under identical illumination conditions: row-sown cotton (low Ω), row-sown soybean (medium Ω), silver birch (*B. pendula*, very low Ω), and European ash (*F. excelsior*, very low Ω) at 4-m crown spacing. All four cases yield positive f_s. The positive value for European ash despite its near-erectophile LAD ($\cos\theta = 0.34$) confirms that crown-level spatial aggregation, not leaf angle, is the governing variable.

even in the absence of specular reflection. Second, because the reflectance factor is normalized by irradiance on a horizontal reference plane while individual leaves are tilted, LAD-driven cosine-projection effects can create a mismatch between the normalization reference and the local leaf irradiance, making TOC reflectance increasingly decoupled from intrinsic leaf reflectance at centimeter resolution and yielding apparent reflectance values exceeding 1 for some fine-scale pixels.

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High-SZA cases are used here to probe method robustness rather than to suggest preferred acquisition geometries. This effect becomes more evident at centimeter-level resolution, where individual leaf facets are partially resolved, and their orientations strongly modulate the local illumination relative to the horizontal-plane normalization.

At the same time, high SZAs combined with varied leaf inclinations lead to many leaves being illuminated on their abaxial (back) surface. This increases the transmitted component captured by the sensor, especially in the NIR where leaf transmittance is high, adding to the upwelling signal even though its pathway differs from an idealized surface-reflection interpretation. The high apparent reflectance values in certain pixels [see Fig. 3(g)–(i)] are therefore a composite of orientation-driven illumination and enhanced transmission through underside-illuminated leaves. LESS simulations that exclude specular reflection still show this tendency, underscoring the structural-illumination pathways involved, while in field measurements, geometry-driven anisotropy and potentially enhanced specular contributions may further contribute under high SZAs.

Canopy structure is hierarchical, and different levels (leaf, branch, plant, and population) influence apparent reflectance as solar angle changes. In row-planted canopies, mutual shading among plants intensifies at high SZAs, creating larger shadows. Consistently with Fig. 4, the proportion of the lower-mean-reflectance component (component 1) increases and the higher-mean-reflectance component (component 2) decreases as SZA rises, reflecting strengthened mutual shading under oblique illumination. Under overcast conditions, directional effects weaken, and the reflectance distribution may respond differently to SZA. We also observed a higher fraction of reflectance >1 at 900 nm under high SZAs, as noted in prior studies [65], [66], potentially involving multiple factors including specular reflection.

Taken together, these mechanisms imply that under high SZAs the TOC reflectance distribution is shaped by multiple within-canopy light pathways (orientation-driven illumination, enhanced transmission, and mutual shading), which naturally produces skewness and heavier tails rather than a simple bimodal pattern. In such cases, enforcing a dichotomous partition (sunlit versus shaded) or fitting the distribution with only two Gaussian components can be insufficient to represent the full reflectance probability density function (PDF), especially for accurately characterizing the upper tail where high-reflectance pixels become more frequent [see Figs. 3(g) and (h) and 4(d)–(f)]. This distributional complexity provides a direct methodological motivation for increasing the number of GMM components (e.g., $K \geq 3$) to better approximate the continuous reflectance distribution and more consistently isolate the most representative illumination-related reflectance component for subsequent retrieval.

Both high- and low-mean-reflectance components carry useful but different information. Low-mean-reflectance components, dominated by scattered and transmitted light, can alter spectral shape and normalization, and may therefore bias LCC retrieval high in some cases. High-mean-reflectance components correspond to facets directly illuminated by sunlight; when not dominated by extreme geometric effects (e.g., hotspot), their apparent reflectance is more stable and tends to yield more accurate LCC retrievals.

In an idealized, perfectly Lambertian and flat canopy, apparent reflectance would remain constant with SZA and always be the highest for the sunlit portion. In real canopies, we use the GMM to isolate the high-mean-apparent-reflectance pixels that best retain leaf physiological and biochemical information. The consistency of improved LCC estimates across the 2022 and 2023 datasets likely reflects GMM's ability to separate illumination heterogeneity rather than year-specific flight geometry. Related simulations indicate that under high canopy cover (e.g., $f_{\text{Cover}} > 0.83$), the impact of row structure on apparent reflectance becomes less pronounced, further supporting the stability of these results.

In some acquisition geometries, hotspot effects may bias the highest-mean-reflectance component, so it is not always the most reliable. For the cotton canopies studied here, drought and salt stress can reduce plant height and leaf number, decreasing the proportion of high-mean-reflectance components at high SZAs. Analyzing the probability distribution of within-canopy apparent reflectance may provide complementary evidence of stress beyond LCC estimation.

High SZA conditions are often unavoidable at higher latitudes and during wintertime acquisitions, where even the solar-noon SZA can remain above 45°, and UAV observations must remain robust under such nonideal geometries. Treating high SZA as a stress test highlights how componentwise or distribution-based approaches can help: they temper the influence of heavy tails and geometric extremes while retaining the informative high-mean-reflectance component for retrieval.

D. Limitations and Prospects

This study advances the characterization of within-canopy reflectance variability and chlorophyll estimation under varying illumination. Several limitations should be noted when interpreting the generality of the conclusions.

First, the spatial resolution of the measured data is 4×4 cm per pixel, which is sufficient to resolve leaf-scale light and shade patterns in the target cotton canopies. While finer spatial resolution can theoretically isolate smaller canopy subcomponents, centimeter-level UAV hyperspectral imagery introduces stronger spectral variability and higher processing complexity due to enhanced within-canopy heterogeneity and local illumination effects [76], [77]. The method's principal boundary is spatial resolution: as the pixel size coarsens, each pixel integrates leaf facets at different illumination states, eroding the spatial aggregation structure (see Section IV-B) that produces a separable bright subpopulation; controlled simulations show the CI_{re} f_s declining from +0.89 at 4 cm to +0.64 at 30 cm (see Supplementary Fig. S4). Accordingly, the operational resolution should be empirically verified for the target crop, sensor, and canopy architecture to balance component separability against signal-to-noise ratio. Second, the vertical heterogeneity of the canopy was not explicitly addressed. Our approach primarily utilizes TOC reflectance, which is dominated by the upper leaf layers. In reality, leaf traits and structural properties vary vertically within the

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canopy profile due to light adaptation and growth stages. Neglecting the vertical variation of leaf functional parameters (e.g., LCC gradients) may introduce bias when scaling leaf-level estimates to the canopy level, particularly for dense canopies where lower layers contribute significantly to the total biomass but less to the radiometric signal. Third, the selection of the number of Gaussian components (K) lacks a universal standard; the appropriate value depends on canopy structural complexity, spatial resolution, and illumination conditions, as well as on the number of pixels available. A practical constraint is that the components are unequally populated: the high- and low-reflectance components lie in the tails of the distribution and therefore tend to contain fewer pixels than the central components, so a finer partition progressively reduces the pixel count of the high-reflectance component on which the retrieval relies. This upper bound is more restrictive for the small field plots than for the large-sample simulations, which is why the field LCC analyses adopt a coarser partition ($K = 3$) than the $K = 2$–5 range explored on the simulated data. We evaluated $K = 2$–5 and capped K at 5 to avoid over-fragmentation and maintain interpretability; the main trends were robust for $K = 3$–5. Adaptive methods for determining K per scene, balancing component separability against per-component sample size, could be explored in future work.

Fourth, the proportion of sky-scattered light (*skyl*) was parameterized using representative levels rather than continuous time-series inputs. We measured downwelling irradiance to determine the observed *skyl* range (25%, 50%, and 95%) for the LESS simulations. However, rapid, instantaneous fluctuations in *skyl* magnitude and spectral distribution under intermittent cloud cover were not explicitly modeled. This limitation implies that the method is currently most robust under stable sky conditions. More broadly, acquisition-side dynamics during flight can perturb the quasi-stationary illumination distribution that the decomposition assumes: transient clouds shift the overall irradiance and hence the brightness boundaries between components, while wind-induced canopy motion momentarily alters leaf orientation and mutual shading, blurring the per-pixel apparent-reflectance distribution. These effects are mitigated by acquiring within short, stable weather windows (each flight completed within about 10 min here), synchronizing downwelling-irradiance measurements, and applying frame-level quality control; they are acquisition factors outside the method's control rather than limitations of the decomposition itself. Finally, regarding the calibration, SPAD was used as a proxy for LCC. While showing good correlation, the relationship was calibrated using limited laboratory sampling campaigns. Future work would benefit from more frequent destructive sampling across growth stages to account for stage-dependent variations in the SPAD–chlorophyll relationship.

Based on these limitations, future research could expand this framework in several directions.

1) *Multitrait Estimation:* Extending the model to retrieve additional leaf traits (e.g., water content and dry matter) simultaneously.
2) *Stress Response:* Testing the method's sensitivity to abiotic or biotic stresses (e.g., drought or disease) that alter both canopy structure and illumination mixing.
3) *Temporal Dynamics:* Evaluating diurnal cycles through repeated flights to jointly capture the interaction between time-varying plant physiology and solar geometry.

V. Conclusion

At the centimeter-level spatial resolution of UAV hyperspectral imagery, canopy reflectance exhibits pronounced within-canopy variability driven by directional illumination and 3-D canopy structure. Under such conditions, treating canopy reflectance as a spatially homogeneous signal or enforcing a strict sunlit–shaded dichotomy often leads to biased or unstable estimates of LCC.

This study demonstrates that explicitly representing the continuous distribution of within-canopy reflectance and isolating illumination-consistent reflectance components can improve LCC retrieval. Using a probabilistic, pixelwise component assignment framework, the highest-mean-reflectance component consistently yielded more accurate and stable estimates across different retrieval paradigms, including index-based regression, physics-based radiative transfer inversion, and data-driven multivariate modeling. The magnitude and persistence of these improvements under varying SZAs indicate that the observed gains arise from improved reflectance representativeness rather than from any specific retrieval model or inversion strategy.

More broadly, these findings suggest that improving the physical representativeness of reflectance at the pixel-distribution level is a critical step toward robust retrieval of vegetation biochemical traits from high-resolution UAV imagery. Distribution-aware reflectance representations therefore offer a promising pathway for enhancing the transferability and stability of trait estimation across sensors, acquisition geometries, and modeling frameworks.

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